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MSCI RESEARCH PROJECT

Variational Methods for Option Pricing

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Abstract

Type your abstract here. The abstract is a summary of the contents of the project. It should be brief but informative, and should avoid technicalities as far as possible.

Acknowledgments

Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

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Chapter 1

Introduction

1.1 Motivation

The financial market offers a much wider variety of financial instruments than just stocks and bonds. Derivatives are financial instruments whose value depends on the value of an underlying asset, which could be a stock, a bond, an index or even another derivative. These are useful in the real world for hedging and speculation, widely used by institutions and individuals alike, and are mathematically rich objects that have been the subject of intense research for decades.

Options are a specific type of derivative contract that gives the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined strike price K , specified at the time of option purchase. These contracts have a fixed expiry time T , at which the holder must decide whether to exercise the option or let it expire worthless. Different types of options exist: European options can only be exercised at the expiry time T , while American options can be exercised at any time before expiration. With the increased flexibility of American options, they are more widely used in practice, and are more interesting mathematical objects to study, since they are much more difficult to price. Holders of this contract must ask themselves constantly whether they should settle for the current value by exercising the option immediately, or wait for a potentially better opportunity.

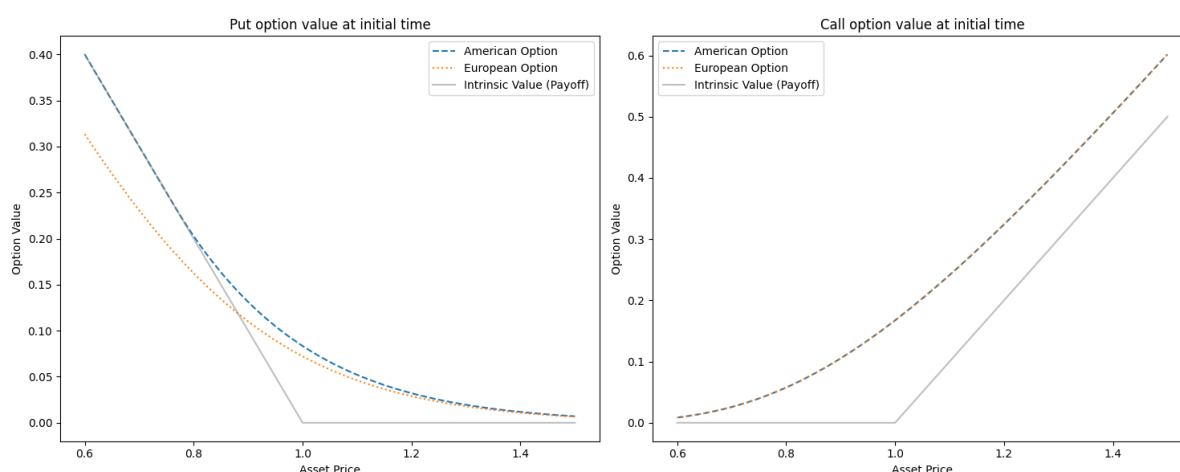


Figure 1.1 Option values (for put and call) at initial time

In this project, we will concentrate our attention towards put options (which give the holder the right to sell the underlying asset), since for call options (which give the holder the right to

buy the underlying asset), the American option is equivalent to the European option if there are no dividends, which we will assume for simplicity. At expiry, the payoff of the option is given by $\max(K - S_T, 0)$, where S_T is the price of the underlying asset at time T . This is because holders of the put option should sell when the price of the underlying asset is below the strike price. Similarly, for a call option, holders should buy when the price $S_T > K$, and hence the payoff is given by $\max(S_T - K, 0)$. In these cases, the option is in-the-money (expires with a positive payoff). Else, the option is out-of-the-money (expires worthless). Figure 1.1 illustrate these payoff functions at expiry.

Figure 1.1 also shows the value of a put and a call option at the initial time, which can be decomposed into the intrinsic value (the payoff should it be exercised immediately) and the time value (the additional value from the possibility of future price movements). We can see clearly that the American option bounds the European option from above, since the holder of the American option has more flexibility in choosing the exercise time. Interestingly, we can see that the value of the European put is lower than the intrinsic value for in-the-money asset prices, which is a feature of the positive interest rate coupled with the fact the option can only be exercised at expiry. Of course, the intrinsic value bounds the American option from below, since the holder can always choose to exercise immediately.

These options are very important financial instruments, and tens of millions of contracts are traded every day just for the S&P 500 index alone. For that reason, it is important to be able to price these options accurately. This is a non-trivial task, as the price of an option depends on many factors, such as the underlying asset price which is generally modelled as a stochastic process.

1.2 Literature review

The pioneering work in option pricing was done by Black and Scholes [1], who derived a closed-form solution for the price of the European call option under the assumption that the underlying asset price follows a geometric Brownian motion. This model, known as the Black-Scholes model, is considered the foundational work in option pricing theory, and the authors were awarded the Nobel Prize in Economics in 1997 for their contribution. However, it relies on several assumptions such as constant risk-free rate and volatility, which is not always realistic in practice, with real markets showing implied volatility varying across different strike prices and maturities, a phenomenon known as the volatility smile. As a result, people started developing more sophisticated models that took inspiration from the Black-Scholes model. Heston [2] allowed volatility to be its own stochastic process, which is more consistent with empirical observations of financial markets, and derived the European option prices of this proposed model. This allowed for a better fit to the volatility smile and was more flexible in capturing market dynamics.

While the European option has an analytical solution specific models, the American option is much more difficult to price, since it allows the holder to exercise the option at any time before expiration, making it analogous to an optimal stopping problem. These optimisation problems do not admit a closed-form solution in general. As a result, various numerical methods have been developed to tackle this problem, such as the binomial tree method [3, 4] for Black-Scholes and Heston respectively, where the asset price is discretised into a tree structure and the option prices is computed via backward induction from the terminal payoff at expiry. Monte-Carlo simulation methods were also explored [5], where heavy research has gone into optimising the idea of pricing an option by simulating an ensemble of paths and computing the average value. More recently, variational methods are explored, where the optimal stopping problem is formulated as a variational inequality, with deep neural networks finding approximations to the

solution [6, 7], borrowing from the field of physics-informed machine learning. This method is adapted to incorporate fractional Sobolev norms to guarantee convergence of the solution to the true option price instead of just pointwise convergence, which has been particularly developed for American option pricing [8].

1.3 Overview of project

This project aims to numerically solve the American option pricing problem under the Black-Scholes and Heston models on increasingly complex payoffs by iteratively building benchmarks for the simpler cases.

For each individual model, we begin by deriving the price of a European call option, which serves as a benchmark for the binomial tree method. These formulae are well-known and can be found in the original papers [1, 2] but we provide a condensed derivation for completeness, with clearer clarification on the results employed in the steps in the derivation.

The binomial trees can easily be adapted to price American options, which provide a benchmark for the variational neural network. The binomial tree method is a well-established numerical technique for pricing options, but I take an interpolation approach, as suggested in [4], for the Heston model where there are two concurrent processes (the asset and the volatility) to build a tree for. The original method yields good results, but I further adapt the method to allow efficient pricing across the whole pricing domain.

The neural network method is implemented with original loss functions inspired by the PINN framework based on the boundary behaviour of the option price, with weights tuned by adaptive weighting schemes. We additionally implement an alternative loss function based on the fractional Sobolev norm, an original research field explored in [8] by one of my supervisor's PhD students, which I adapt to price American options under the whole domain space instead of just at-the-money as in the original paper.

We use both the neural network and the binomial tree method as benchmarks for the multi-asset case, where we price put options on the product of assets, which has yet to be explored in the literature in depth. This is motivated by the fact that there exists an equivalent single-asset reduction for this payoff, which allows us to validate the results against the benchmarks already established in the one-asset case. With this foundation established, we are then free to tackle alternative payoffs where such a reduction does not exist, such as the maximum or minimum of several assets, and still have confidence in the results since the methods have been validated on the simpler cases.

We will conclude our numerical results by applying our pricing scheme to two further objectives: estimating the free boundary, and estimating the Greeks to gain insight into exposures and hedging strategies. The free boundary is the critical asset price at which the holder of the option is indifferent between exercising immediately or waiting, which I compute numerically through a binary search method for deterministic methods such as the binomial tree. For non-deterministic methods, I propose a novel method to frame the free boundary problem as a soft-classification problem, where I make distributional assumptions on our continuation values and use the empirical variance of the continuation values as a measure of uncertainty that tunes the soft classification. The continuation values are estimated through Monte-Carlo simulation, and the soft classification is done through an empirically tuned injective function.

The project aims to provide a comprehensive numerical framework for pricing American options under two different models, with detailed numerical results highlighting the strengths and shortcomings of each method, and insights into the option price behaviour under different

payoffs, complexity and market conditions. All of these are backed by a rigorous mathematical framework that borrows from the fields of stochastic calculus, numerical analysis and machine learning, allowing a translation of theoretical insights into practical numerical methods that can be applied in real-world scenarios.

Chapter 2

Mathematical Background

2.1 Preliminaries

We will first assemble the necessary mathematical background to understand the option pricing problem and build some foundations into the methods we will be using to solve it. The definitions and results are mostly standard, and can be found in textbooks such as [9, 10] but we will cover them here for self-containment and to set up notation for the rest of the project. As a result, we will only cover the main concepts and not delve too deeply into the details, since our focus is more on the numerical methods and their applications.

2.1.1 Probabilistic Setup

The pricing problem is inherently probabilistic, as we are uncertain about the future evolution of the underlying asset price, thus requiring a rigorous mathematical framework to model this uncertainty. The formalisation of the underlying randomness in our model for the market will be done through probability spaces, and the arrival of information in the market through time through filtrations.

Definition 2.1.1 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is the sample space, $\mathcal{F}$ is a σ -algebra on Ω , and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure such that $\mathbb{P}(\Omega) = 1$.

Definition 2.1.2 (Filtration, filtered probability space). A *filtration* on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a non-decreasing family $(\mathcal{F}_t)_{t \geq 0}$ of sub- σ -algebras of $\mathcal{F}$, that is to say, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for all $s \leq t$. The quadruple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ is called a *filtered probability space*.

We will also define a Martingale, which formalises the idea that future expectations of a process are equal to the current value, which is a key concept in the risk-neutral pricing framework.

Definition 2.1.3 (Martingale). A process $M = (M_t)_{t \geq 0}$ is a *martingale* with respect to the filtration $\{\mathcal{F}_t\}$ and the probability measure $\mathbb{P}$ if $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

The Markov assumption is also commonly made in financial modelling, which states that the future evolution of the process only depends on the current state and not on the past history. It is useful because it greatly simplifies models, and hence we will employ it in our project as well.

Definition 2.1.4 (Markov process). A process $X = (X_t)_{t \geq 0}$ is a *Markov process* with respect to the filtration $\{\mathcal{F}_t\}$ if

$$\mathbb{E}[f(X_t) | \mathcal{F}_s] = \mathbb{E}[f(X_t) | X_s] \quad (2.1)$$

for all $0 \leq s \leq t$ and all bounded measurable functions f .

2.1.2 Stochastic processes

The market in practice is not deterministic, so we will mathematically construct our model with stochastic processes, indexed by time in our application.

Definition 2.1.5 (Stochastic process). A *stochastic process* is a family of random variables $\{X_t : t \geq 0\}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

More specifically, we will use Brownian motion to model the underlying asset price dynamics, which is a continuous-time stochastic process with continuous paths and stationary independent increments.

Definition 2.1.6 (Brownian motion). A real-valued process $W = (W_t)_{t \geq 0}$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a *Brownian motion* if it satisfies the following properties:

- $W_0 = 0$ almost surely
- The increment $W_t - W_s$ is normally distributed with mean 0 and variance $t - s$ for all $0 \leq s < t$
- The increments $W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent for all $0 \leq t_0 < t_1 < \dots < t_n$
- The sample paths $t \mapsto W_t(\omega)$ are continuous

The existence of this process is established by Wiener [11] and is very powerful in modelling the randomness in the financial market.

2.1.3 Risk-neutral pricing

We will now formalise the financial market and establish the principle of arbitrage-free valuation that will underpin our option pricing problem.

In our model, the market involves a fixed horizon $T > 0$ (our expiry) and a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$ where $\{\mathcal{F}_t\}$ is the filtration generated by the underlying Brownian motion. The market consists of a risk-free asset B_t that is deterministic and follows the evolution

$$B_t = \exp\left(\int_0^t r(s) ds\right) \quad (2.2)$$

and one or more risky assets S_t that follows an Itô process.

It is important that the market is arbitrage-free, which means that there does not exist any trading strategy that can generate a positive profit with zero initial investment and zero risk. This trading strategy ϕ must be self-financing, which means that any change in the value of the portfolio is only due to changes in the assets held, without any external cash flow.

Definition 2.1.7 (Self-financing trading strategy). A *self-financing trading strategy* ϕ is a predictable process such that the value of the portfolio $V_t = \phi_t^B B_t + \phi_t^S S_t$ satisfies

$$dV_t = \phi_t^B dB_t + \phi_t^S dS_t \quad (2.3)$$

Definition 2.1.8 (Arbitrage). An *arbitrage* is some trading strategy ϕ such that $V_0 = 0$, $V_T \geq 0$ almost surely and $\mathbb{P}(V_T > 0) > 0$.

By the fundamental theorem of asset pricing, the market is arbitrage-free if and only if there exists a measure $\mathbb{Q}$ such that $\mathbb{Q}$ is equivalent to $\mathbb{P}$ (they have the same null sets) and the discounted price process $\tilde{S}_t = \frac{S_t}{B_t}$ is a martingale under $\mathbb{Q}$. We will assume that our market is complete (any derivative can be replicated), and hence such a measure $\mathbb{Q}$ will be unique.

Given that the market is arbitrage-free, we can price any derivative with payoff $g(\cdot)$ at time t by taking the discounted expectation of the payoff under the risk-neutral measure $\mathbb{Q}$, giving us the risk-neutral pricing formula for European derivative with maturity T :

$$v(t, S) = B_t \mathbb{E}^{\mathbb{Q}} \left[\frac{g(S_T)}{B_T} \mid \mathcal{F}_t \right] \quad (2.4)$$

Throughout this project specifically, we will deal with constant interest rates, so the bond price process reduces to $B_t = e^{rt}$ where r is the constant risk-free rate. Additionally, we will also assume the Markov property (2.1.4). The arbitrage-free price of the European derivative will therefore be given by

$$v(t, S) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} g(S_T) \mid S_t = S \right] \quad (2.5)$$

2.1.4 Optimal stopping problem and free boundary

The American option gives the holder the right to exercise the option at any time before maturity, which thus gives us the interpretation of the option price as the solution to an optimal stopping problem, as we need to additionally identify when we should exercise the option.

Definition 2.1.9 (Stopping time). A random variable $\tau : \Omega \rightarrow [0, \infty]$ is a *stopping time* if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

The price of the American option can thus be expressed as the $\mathbb{Q}$ -expectation of the discounted payoff at the optimal stopping time τ^* that maximises the expected payoff, giving the optimal stopping problem

$$v(t, S) = \sup_{\tau \in [t, T]} \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau-t)} g(S_\tau) \mid S_t = S \right] \quad (2.6)$$

which is simply (2.5) with the additional optimisation over the stopping time τ .

This optimal τ^* is the first time that the option holder chooses to exercise the option, which motivates the notion of a free boundary, which is the boundary separating the exercise region and the continuation region.

Let's work with the case $t = 0$ for simplicity, but the same logic applies for any $t \in [0, T]$. For the American put, the optimal stopping problem (2.6) can be rewritten as

$$v(0, S) = \sup_{\tau \in [0, T]} \mathbb{E} \left[e^{-r\tau} (K - S_\tau)^+ \mid S_0 = S \right] \quad (2.7)$$

which is achieved by an optimal stopping time τ^* taking the form

$$\tau^* = \inf \{ \tau \geq 0 : S_\tau \leq b^*(\tau) \} \quad (2.8)$$

for some optimal exercise boundary $b^*(\tau)$ [5]. This is illustrated in Figure 2.1, where we simulate one realisation of the asset price assuming a geometric Brownian motion and plot it against the exercise boundary. The illustration is based on the binomial tree solution on the American put option, which has the boundary

$$b^*(\tau) = \sup \{ S : c(\tau, S) = g(S) \} \quad (2.9)$$

which is found using binary search here, and $c(t, S)$ is the continuation value of the option with payoff $g(S)$.

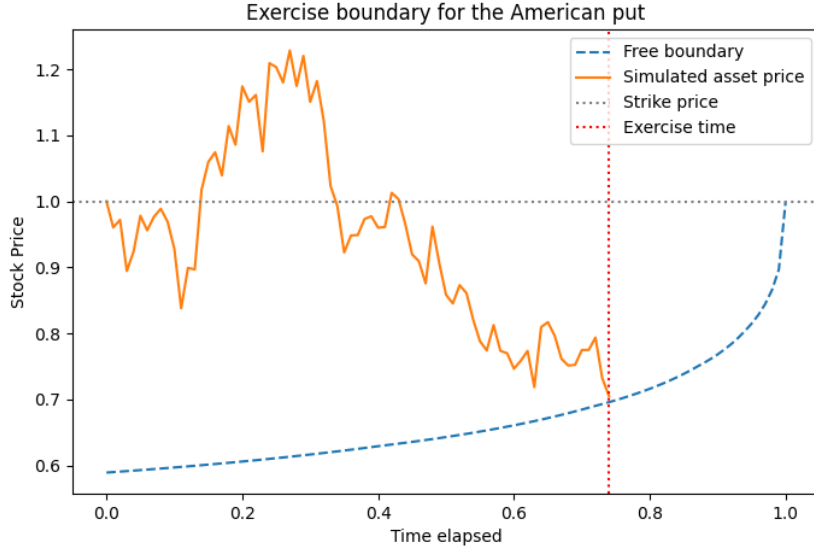


Figure 2.1 Exercise boundary for the American put (adapted from [5])

Soft classification of the free boundary

This formulation describes a hard classification problem, where we classify points in the time-stock price space binarily to be either in the continuation region or the exercise region. However, our numerical solutions to the pricing problem are not exact and there is some level of error in the solution that comes from the Monte Carlo simulations upon the space, the numerical differentiation used in training and the early stopping in the training process. Therefore, it would be more intuitive to perform a soft classification instead, where we assign probabilities to points in the time-stock price space to be in the continuation region or the exercise region, which would give us a more intuitive visualisation of the free boundary and how it evolves over time. We will do this by extending hard classification: that is to say, the continuation region A is

$$A = \{(t, S) : c(t, S) > g(S)\} \cup \{(t, S) : g(S) = 0\} \quad (2.10)$$

where intuitively, the first set is those where the continuation value is greater than the early exercise value (and hence it is better to continue), and the second set is those where the payoff is zero, which are at-the-money and out-of-the-money options that are worthless under immediate exercise.

Sigmoid soft classification

We will consider two methods to extend the definition of the continuation region to a soft classification. The first one is to use a sigmoid function $s : \mathbb{R} \rightarrow (0, 1)$ defined as

$$s(x) = \frac{1}{1 + e^{-x}} \quad (2.11)$$

to smoothly transition the continuation probability from 0 to 1 as the value function transitions from being below the payoff to being above the payoff. As the sigmoid function is the CDF of the logistic distribution, we can interpret this as modelling the noise in the value function as a logistic distribution. The effect of the noise can be controlled by introducing the softness parameter $\tau > 0$, and our continuation probability function can be defined on the relative

difference from the payoff as

$$P_{\text{cont}}(d) = s \left(\frac{d}{\tau} \right), \quad d := \frac{c(t, S) - g(S)}{g(S)} \quad (2.12)$$

To tune the softness parameter, we can define a threshold $I \in (0.5, 1)$ such that we want the continuation probability to be I at some relative difference $\epsilon > 0$, i.e. $P_{\text{cont}}(\epsilon) = I$, which we could solve with some simple algebraic manipulation to get

$$\tau = \frac{\epsilon}{\ln \left(\frac{I}{1-I} \right)} \quad (2.13)$$

Gaussian noise soft classification

The second method is to assume Gaussianity of the noise in our estimate $\hat{c}$ of the continuation value, and that our approximated price is unbiased. In other words, we have

$$\hat{c}(t, S) = c(t, S) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \tilde{\sigma}^2) \quad (2.14)$$

where c is the true (latent) continuation value and $\tilde{\sigma}$ is the standard deviation of the solution noise, which is also unknown. We can then perform solve the problem n times to get the different approximated continuation values $\hat{c}_1, \dots, \hat{c}_n$. This allows us to use the sample mean $\bar{c}$ as our unbiased estimate of the continuation value function paired with the sample standard deviation $\hat{\sigma}$ as our estimate. Then $\bar{c}$ is distributed according to a t -distribution with $n-1$ degrees of freedom, and we can compute the continuation probability as

$$P_{\text{cont}}(t, S) = \mathbb{P}_{t_{n-1}} \left(\frac{\bar{c}(t, S) - g(S)}{\hat{\sigma}(t, S)/\sqrt{n}} > 0 \right) \quad (2.15)$$

which is simple as the CDF of the t -distribution is readily available.

The caveat with this method is that it requires multiple simulations of the solution, which becomes infeasible for more complex models and higher dimensions, but it gives us a more statistically grounded way to compute the continuation probability.

All that is left is to find a method to obtain the continuation value function, as it is generally latent and not directly observable, given that we predict the American option price that already takes the maximum of the continuation value and the payoff. However, we can use

$$c(t, S) = \mathbb{E} \left[e^{-rh} f(t+h, S_{t+h}) \mid S_t = S \right] \quad (2.16)$$

where $h > 0$ is a small time step and f is the American option price function. We can obtain an unbiased estimation of this expectation by Monte Carlo: we simulate the asset price S_{t+h} at time $t+h$ given $S_t = S$ many times and evaluate the option price function for each simulation, then taking the mean.

2.1.5 Monte Carlo integration

Monte Carlo is a powerful way to estimate integrals that may be intractable or not have a closed-form solution, which will be useful through our project as expectations can be expressed as integrals under the probability measure. The idea is to use the law of large numbers to estimate the integral by sampling from some distribution and evaluating the integrand at the sampled points.

Suppose we want to compute the integral of some function ϕ over a domain Ω :

$$I = \int_{\Omega} \phi(x) \, dx \quad (2.17)$$

If we are able to sample iid from some distribution with probability density function q that has positive density on Ω , then we can rewrite the integral as

$$I = \int_{\Omega} \frac{\phi(x)}{q(x)} q(x) \, dx = \mathbb{E} \left[\frac{\phi(X)}{q(X)} \right] \quad (2.18)$$

where the expectation is taken under the distribution with density q . This expectation can be estimated by sampling $X_1, \dots, X_N$ iid from q and taking the sample mean:

$$\hat{I} = \frac{1}{N} \sum_{j=1}^N \frac{\phi(X_j)}{q(X_j)} \quad (2.19)$$

This is known as importance sampling [12], and our choice of q can affect the variance of the estimator $\hat{I}$. In practice, we can choose q to be uniform over Ω , giving us the simple estimator

$$\hat{I} = \frac{1}{N} \sum_{j=1}^N \phi(X_j) \quad (2.20)$$

2.2 State Space Discretisation

2.2.1 Background

Recall that the option value function $v(t, S)$ is the solution to the optimal stopping problem

$$v(t, S) = \sup_{\tau \in [t, T]} \mathbb{E} \left[e^{-r(\tau-t)} g(S_{\tau}) \mid S_t = S \right] \quad (2.21)$$

There does not exist a closed-form solution for this problem in general, and thus we have to resort to numerical methods. We first consider methods that discretise the continuous time into a finite grid $\{t_i\}_{i=1}^N$ where we have

$$0 = t_0 < t_1 < \dots < t_N = T \quad (2.22)$$

which we can use to index the underlying state variable S_{t_i} at time t_i . Note that the state variable S_{t_i} could depend on multiple variables, such as in the Heston model, where we have a stochastic variance term encoded in the asset price. The goal is to compute the arbitrage-free price $v(0, S_0)$ at time $t = 0$ given the initial state S_0 . As we know the solution must satisfy the terminal condition $v(T, S) = g(S)$ for all S , we can use backward induction to compute the option value at each time step. This gives rise to the class of numerical methods known as tree methods.

2.2.2 Tree Methods

In a tree model, we discretise the time-state space into a finite grid, where the state variable S_{t_i} can take a finite number of values at each time step given the previous one. We will particularly explore binomial trees in this project, where each state can evolve to one of two possible states

at the next time step. Formally, suppose at time t_n we have the state $S_{t_n} \in \{S_n^1, \dots, S_n^{M_n}\}$ where M_n is the number of possible states at time t_n . Then at time t_{n+1} , each state S_n^i can evolve to the state S_{n+1}^j with transition probability

$$p_{ij}^{(n)} = \mathbb{P}(S_{n+1} = S_{n+1}^j \mid S_n = S_n^i) \quad (2.23)$$

This is a discrete-time Markov chain, and to identify the transition probabilities, we can use the risk-neutral condition which states that the discounted price process must be a martingale under the risk-neutral measure. This gives us the condition

$$\sum_{j=1}^{M_{n+1}} p_{ij}^{(n)} S_{n+1}^j = e^{r\Delta t} S_n^i \quad (2.24)$$

where $\Delta t = t_{n+1} - t_n$ and r is the risk-free rate.

With these transition probabilities, we can compute the option value at time t_n given the option value at time t_{n+1} by using backwards induction, which gives us the recursive relation

$$v(t_n, S_n^i) = \max \left(g(S_n^i), e^{-r\Delta t} \sum_{j=1}^{M_{n+1}} p_{ij}^{(n)} v(t_{n+1}, S_{n+1}^j) \right) \quad (2.25)$$

where we take the maximum of the payoff and the discounted expected future price to account for early exercise for the American option. This maximum operator can easily be lifted for all time steps for European options, or only imposed on certain time steps for Bermudan options, leaving the tree method a very flexible method. This backwards induction is applied iteratively starting from the terminal condition at time $t_N = T$ to compute the option value at time $t_0 = 0$.

An issue with tree methods is that the number of states M_n can grow exponentially with the number of time steps N , leading to a very computationally expensive algorithm. To mitigate this, we use recombining trees where the states at each time step are designed to recombine with each other, so that the number of states M_n grows linearly. Another issue is that, while flexible, the tree method can be difficult to implement for more complex models where the state depends on multiple variables or if we wish to extend it to options operating on multiple assets, but its reliability and interpretability make it a good benchmark for simpler models.

2.3 Physics-Informed Neural Networks

2.3.1 Motivation

Our option pricing problem can alternatively be formulated as a variational inequality which takes the form of a partial differential equation that is independent from the payoff $g(\cdot)$. These PDEs can be of high dimension, and are thus difficult to solve using traditional numerical methods due to the curse of dimensionality. Traditional methods such as finite difference methods require discretisation of the domain and the number of grid points (and hence algorithm complexity) scales exponentially $\mathcal{O}(N^d)$ with the dimension d of the problem.

We can attempt a neural network solution to the PDEs, as neural networks are known to be universal approximators. That is, suppose we have the class of functions $\mathcal{F}$ over a compact set S that can be represented by a neural network with a given architecture, with a continuous target function f^* that we would like to approximate. Then for any target accuracy $\epsilon > 0$, there exists a function $f \in \mathcal{F}$ such that

$$\sup_{x \in S} |f(x) - f^*(x)| \leq \epsilon \quad (2.26)$$

If we have a squashing activation function $\Psi(\cdot)$ that is non-decreasing and satisfies the limits

$$\Psi(-\infty) = 0, \quad \Psi(\infty) = 1 \quad (2.27)$$

then the class of functions $\mathcal{F}$ that can be represented by a neural network with one hidden layer and activation function Ψ is dense in the space of continuous functions on S and thus is a universal approximator [13].

Since the option price function is continuous in the time-asset price domain, we can use a neural network to find an approximation. This is favourable because the time complexity scales with the number of parameters, which is not exponential. To do this, we will use the framework of physics-informed neural networks (PINNs) to solve the PDEs, which have been shown to be effective for solving high-dimensional PDEs [6], by avoiding mesh construction and leveraging the interpolation capability of neural networks [14]. The loss function of the neural network is designed to penalise deviations from the PDE as well as deviations from the boundary conditions. By training the neural network to minimise this loss function, we can obtain an approximation to the solution of the PDE. This method has been applied to option pricing problems in the literature, such as in [15] for the Black-Scholes model, but we will look to apply it to more complex models such as the Heston model as well as to options on multiple assets.

2.3.2 Methodology

We first provide a general setup for PINNs, before modifying it to tailor our specific problem of option pricing. Formally, suppose we have an unknown latent solution $u(t, x)$ which is governed by a PDE of the form

$$\frac{\partial u}{\partial t} + \mathcal{L}[u] = 0, \quad t \in [0, T], x \in \Omega \quad (2.28)$$

where $\mathcal{L}$ is a (not necessarily linear) differential operator. Suppose additionally that this PDE is subject to the boundary conditions

$$u(0, x) = g(x), \quad x \in \Omega \quad (2.29)$$

$$\mathcal{B}[u] = 0, \quad t \in [0, T], x \in \partial\Omega \quad (2.30)$$

where $\mathcal{B}$ is a boundary operator. Then the unknown solution u can be approximated by a deep neural network $f_\theta(t, x)$ where θ are the parameters of the neural network, trained by minimising a loss function composed of deviations from the PDE residual, boundary conditions and initial conditions [16]:

$$L(\theta) = \lambda_r L_r(\theta) + \lambda_{bc} L_{bc}(\theta) + \lambda_{ic} L_{ic}(\theta) \quad (2.31)$$

where the loss components are mathematically defined as

$$L_r(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} |\mathcal{R}_\theta(t_r^i, x_r^i)|^2 \quad (2.32)$$

$$L_{bc}(\theta) = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} |\mathcal{B}[f_\theta](t_{bc}^i, x_{bc}^i)|^2 \quad (2.33)$$

$$L_{ic}(\theta) = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} |f_\theta(0, x_{ic}^i) - g(x_{ic}^i)|^2 \quad (2.34)$$

where the PDE residual $\mathcal{R}_\theta$ is defined as

$$\mathcal{R}_\theta(t, x) = \frac{\partial f_\theta}{\partial t}(t, x) + \mathcal{L}[f_\theta](t, x) \quad (2.35)$$

We additionally use the sampled points $\{x_{ic}^i\}_{i=1}^{N_{ic}}$ from the initial condition domain, $\{(t_{bc}^i, x_{bc}^i)\}_{i=1}^{N_{bc}}$ from the boundary condition domain and $\{(t_r^i, x_r^i)\}_{i=1}^{N_r}$ from the PDE domain, sampled uniformly in their respective domains at each iteration of the gradient descent algorithm.

There are different methods to choose the loss weights $\lambda_r, \lambda_{bc}, \lambda_{ic}$, such as using fixed weights or using adaptive weights that change during training, which we will discuss in Section 2.3.3.

For our specific problem of option pricing, we can modify the above setup to fit our variational inequality formulation. The PDE to solve is dependent on the model for the asset dynamics, and the boundary conditions are given by the payoff function of the option. In general, we will have to solve a problem of the form

$$\min \left\{ -\frac{\partial u}{\partial t} + \mathcal{L}[u], u - g \right\} = 0, \quad t \in [0, T], x \in \Omega \quad (2.36)$$

$$u(T, x) = g(x), \quad x \in \Omega \quad (2.37)$$

$$\mathcal{B}_k[u] = 0 \quad t \in [0, T], x \in \partial\Omega, k = 1, \dots, K \quad (2.38)$$

where k indexes the K different boundary conditions. This is very similar to the general setup for PINNs, except that we have a variational inequality instead of a PDE, and we have a terminal condition instead of an initial condition. However, we can still use the framework by noting we can make the substitution $t \mapsto T - t$ to un-negate the negative partial derivative in the PDE term and to convert the terminal condition to an initial condition. The variational inequality can be handled by modifying the PDE residual loss (2.35) to

$$\mathcal{R}_\theta(t, x) = \min \left\{ \frac{\partial f_\theta}{\partial t}(t, x) + \mathcal{L}[f_\theta](t, x), f_\theta(t, x) - g(x) \right\} \quad (2.39)$$

With everything in place, we can train the neural network with gradient descent:

$$\theta_{n+1} = \theta_n - \eta_n \nabla_\theta L(\theta_n) \quad (2.40)$$

where η_n is the learning rate at iteration n , either kept constant or decayed over time with a schedule.

2.3.3 Adaptive Loss Weighting

While keeping the loss weights $\lambda_r, \lambda_{bc}, \lambda_{ic}$ fixed is simple, it can be difficult to tune correctly and can lead to suboptimal performance. Ideally, we would like to balance the different loss components so that they are of the same order of magnitude, which can be achieved by using adaptive loss weighting methods, as proposed in [16]. The idea is to evaluate the contribution of each loss component on the total loss and adjust the weights accordingly to balance their contributions. In practice, we will use a moving average to prevent the weights from changing too rapidly, leading to instability in training. Mathematically, we compute the adaptive weights

$$\begin{aligned} \hat{\lambda}_r &= \frac{\|\nabla_\theta L_r(\theta)\| + \|\nabla_\theta L_{bc}(\theta)\| + \|\nabla_\theta L_{ic}(\theta)\|}{\|\nabla_\theta L_r(\theta)\|} \\ \hat{\lambda}_{bc} &= \frac{\|\nabla_\theta L_r(\theta)\| + \|\nabla_\theta L_{bc}(\theta)\| + \|\nabla_\theta L_{ic}(\theta)\|}{\|\nabla_\theta L_{bc}(\theta)\|} \\ \hat{\lambda}_{ic} &= \frac{\|\nabla_\theta L_r(\theta)\| + \|\nabla_\theta L_{bc}(\theta)\| + \|\nabla_\theta L_{ic}(\theta)\|}{\|\nabla_\theta L_{ic}(\theta)\|} \end{aligned}$$

and update the global weights $\lambda = (\lambda_r, \lambda_{bc}, \lambda_{ic})$ with a smoothness $\alpha \in (0, 1)$ as a moving average

$$\lambda_{\text{new}} = \alpha \hat{\lambda}_{\text{old}} + (1 - \alpha) \hat{\lambda}_{\text{new}} \quad (2.41)$$

This is not performed at each iteration, but rather at a fixed frequency to prevent the weights from changing too rapidly. The literature [16] suggests to update the weights every 1000 iterations with $\alpha = 0.9$ in general.

2.3.4 Numerical Optimisation

Gradient Descent

Our neural network will learn through gradient descent, in particular Adam, which uses adaptive learning rates and momentum to speed up convergence. Note this does not conflict with the adaptive loss weighting described in Section 2.3.3, as the adaptive loss weighting is used to balance the different loss components, while Adam is used to adapt the learning rate for each parameter based on the historical gradients.

We will additionally impose a schedule to decay the learning rate over time, which starts at a conventional value of 10^{-3} to allow for fast learning at the beginning, and decays by a factor for every certain number of iterations, allowing for finer convergence in later iterations. These are also tuned by observing the training curve, and I choose to decay the learning rate by a factor of 0.7 every 2000 iterations.

Early Stopping

Early stopping is a regularisation technique mainly used to prevent overfitting in neural network training. However, we are training on a new random training set each epoch, so we do not need to worry about overfitting, and early stopping is implemented as an optimisation procedure.

The idea of early stopping is to stop the training process once we do not observe a significant improvement in the loss function after a certain number of iterations, which we call the patience. This can be tuned by observing the training curve and finding a good balance between training time and performance. Once the patience is exceeded without observing a significant improvement, the training is stopped and the model reverted to the best performing model observed during training. The loss used to evaluate the improvement is a validation set that is held through the training process, a set of points in the time-state space that is not used for training but only for evaluation.

Throughout this project, I will set the improvement threshold to be 10^{-6} and the patience to be 500 iterations, which I found to be a good balance for the models I trained.

Chapter 3

Black-Scholes Model

The Black-Scholes model [1] is the fundamental model for option pricing, which assumes that the underlying asset price follows a geometric Brownian motion. As with the market introduced in section (2.1.3), the market consists of two underlying assets:

- The risk-free asset, described by a deterministic function

$$dB_t = rB_t dt \quad (3.1)$$

where for convenience, we set $B_0 = 1$ and r is the constant risk-free interest rate with continuous compounding. This has the solution $B_t = e^{rt}$ for $t \geq 0$.

- The risky asset S_t , which has the price process of a geometric Brownian motion (GBM), which satisfies the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad 0 \leq t \leq T \quad (3.2)$$

for a given drift μ and volatility σ . We can solve this SDE to yield the unique solution

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right) \quad 0 \leq t \leq T \quad (3.3)$$

Under risk-neutral pricing, we can find an equivalent probability measure $\mathbb{Q}$ under which the discounted stock price is a martingale, and hence all assets earn the risk-free rate in expectation after discounting. Under the risk-neutral measure, the SDE for the risky asset becomes

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}} \quad (3.4)$$

where the drift has been replaced by the risk-free rate r . The derivation can be found in Appendix A.1.

3.1 Closed-form solution for European options

Our first benchmark will be the closed-form solution for the European option, which would be used to evaluate the accuracy of the binomial tree later. The derivation of this formula is widely available and hence I will only include the main ideas. Using the risk-neutral pricing formula, the value of the European call option with strike K and expiry T is given by

$$v(0, S_0) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ | S_0] \quad (3.5)$$

We use the solution of geometric brownian motion (3.3) to note that

$$S_T = S_0 \exp \left((r - \frac{1}{2}\sigma^2)T + \sigma W_T^{\mathbb{Q}} \right) \quad (3.6)$$

under the risk-neutral measure $\mathbb{Q}$, noting we have replaced the drift μ with the risk-free rate r . The Brownian motion is essentially the standard normal variable $Z \sim \mathcal{N}(0, 1)$ so we have

$$S_T = S_0 \exp \left((r - \frac{1}{2}\sigma^2)T + \sigma Z \right) \quad (3.7)$$

This can be plugged into the risk-neutral pricing formula (3.5): the exercise event $\{S_T > K\}$ is equivalent to $\{Z > -d_2\}$ where

$$d_2 = \frac{\log(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_1 = d_2 + \sigma\sqrt{T} \quad (3.8)$$

The expectation (3.5) can be split

$$v(0, S) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \quad (3.9)$$

$$= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)\mathbf{1}_{\{S_T > K\}}] \quad (3.10)$$

$$= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}}] - K e^{-rT} \mathbb{Q}(S_T > K) \quad (3.11)$$

The second probability is immediate as we see $\mathbb{Q}(S_T > K) = \mathbb{Q}(Z > d_2) = \Phi(d_2)$ by symmetry, where Φ is the CDF of the standard normal. We can complete the square in the Gaussian integrand for the first expectation to get the final formula [1]

$$v(0, S_0) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2) \quad (3.12)$$

3.2 Binomial Tree Method

We can use the binomial discrete-time model to price both European and American options with the tree method. This model uses the simplification that the stock price can only move to one of two possible new prices in a time step. To avoid explosion of nodes, we will only consider recombinant trees so that the algorithm performs with quadratic time complexity. We will verify the approximated solution's validity against the closed-form formula for European options, and use its American variant as a benchmark for later methods.

To formalise the tree construction, let's fix a time horizon $T > 0$. For some depth $n \geq 1$, divide the interval $[0, T]$ into n even steps each of length $\Delta t = T/n$. Our asset S_t will follow a multiplicative binomial evolution:

$$S_{i+1} = \begin{cases} S_i u & \text{with probability } p \\ S_i d & \text{with probability } 1 - p \end{cases} \quad S_0 > 0 \quad (3.13)$$

where u and d are the model parameters determining the two factors of the stock price change per time step.

As with (3.1), the financial market will also consist of a risk-free asset B_t , discretised so that it satisfies

$$B_{i+1} = B_i e^{r\Delta t}, \quad B_0 = 1 \quad (3.14)$$

where r is the continuously-compounded risk-free rate, typically considered interest. For an arbitrage free market, we should additionally ensure that

$$d < e^{r\Delta t} < u \quad (3.15)$$

for sufficiently small Δt , else we can make a riskless profit by shorting the stock and investing in the risk-free asset, or vice versa.

To employ risk-neutral pricing, we additionally require the discounted asset price to have zero drift (admitting a martingale measure). More precisely, let the discounted price be

$$\tilde{S}_i = e^{-ri\Delta t} S_i \quad (3.16)$$

We require some probability measure $\mathbb{Q}$, equivalent to the physical measure $\mathbb{P}$ (have the same non-zero support) such that

$$\mathbb{E}_{\mathbb{Q}}[\tilde{S}_{i+1} \mid \mathcal{F}_i] = \tilde{S}_i \quad (3.17)$$

To calculate the risk-neutral probability, we can apply this to our model:

$$e^{-r(i+1)\Delta t} [pS_i u + (1-p)S_i d] = e^{-ri\Delta t} S_i \quad (3.18)$$

Collect the p terms and cancel out the discount factor to get

$$p(S_i u - S_i d) + S_i d = e^{r\Delta t} S_i \quad (3.19)$$

Divide both sides by S_i and rearrange to make p the subject to obtain

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (3.20)$$

This is what we will use in our binomial model.

To price the option at time 0, we will perform backward induction from expiry, since we know that the price is just the payoff $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ as the option expires.

As the discounted price process is a martingale under the risk-free measure $\mathbb{Q}$, the arbitrage-free value $v(i\Delta t)$ of the derivative at time $i\Delta t$ is necessarily

$$v_i = \underbrace{e^{-r(T-i\Delta t)}}_{\text{discount}} \mathbb{E}_{\mathbb{Q}}[g(S_T) \mid \mathcal{F}_i] \quad (3.21)$$

We will utilise the Cox-Ross-Rubinstein (CRR) Model [3] for our implementation, which chooses the parameters

$$u = e^{\sigma\sqrt{\Delta t}} \quad d = e^{-\sigma\sqrt{\Delta t}} \quad (3.22)$$

satisfying the arbitrage free condition, and normalising the local variance of returns to the volatility parameter σ^2 . They also conveniently multiply to 1.

To implement this, we need to first compute the price tree. This would be, for the price at step i after j up-moves (so clearly $i \geq j$)

$$S_{i,j} = S_0 u^j d^{i-j} \quad (3.23)$$

As the payoff is known at maturity, we can simply set

$$v_n(j) = g(S_{n,j}) \quad 0 \leq j \leq n \quad (3.24)$$

and recursively set

$$v_i(j) = e^{-r\Delta t} (p v_{i+1}(j+1) + (1-p) v_{i+1}(j)) \quad (3.25)$$

for $0 \leq j \leq i$. This would be it for European options, but since early exercise is allowed, we will compare this value with the payoff of the price at step i with j up-moves, which is just $f(S_{i,j})$. Thus the value is given by

$$v_i(j) = \max \left(e^{-r\Delta t} (p v_{i+1}(j+1) + (1-p) v_{i+1}(j)), g(S_{i,j}) \right) \quad (3.26)$$

The root value $v_0(0)$ would be the arbitrage-free option price.

3.3 Physics-Informed Neural Networks

In order to use the framework of PINNs as detailed in section 2.3, we need to first derive the PDE satisfied by the option price, as well as the boundary conditions. We will first work with the one-dimensional case, then discuss the extension to the multi-dimensional case and how we could build a benchmark to against which we evaluate our results.

3.3.1 PINN formulation for one-dimensional case

In order to set up our PINN, we need to first derive the PDE satisfied by the option price. Recall the value function $v : [0, T] \times \mathbb{R}^+ \rightarrow \mathbb{R}$ of an American put with payoff $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ at time t with asset price $S_t = S$ is given by

$$v(t, S) = \sup_{\tau \in [t, T]} \mathbb{E}_t [e^{-r(\tau-t)} g(S_\tau) \mid S_t = S] \quad (3.27)$$

where τ is any stopping time in $[t, T]$ and $\mathbb{E}_t$ is conditional expectation at time t , given $S_t = S$. For an American option, we always have two options:

- Immediately exercise the option and get payoff $g(S)$. Since this is not necessarily optimal, the payoff must bound the value option from below:

$$v(t, S) \geq g(S) \quad (3.28)$$

- Do nothing and let the option continue to time $t + h$ with $h > 0$ a very small number. By using the supremum definition of $v(t, S)$ (3.27) and swapping the expectation and supremum [17] we arrive at

$$e^{rh} v(t, S) \geq \mathbb{E}_t [v(t + h, S_{t+h})] \quad (3.29)$$

and the inequality turns into an equality at the limit $\delta \rightarrow 0$.

We will additionally assume that the value function $v \in C^{1,2}$ (continuously differentiable with respect to t and twice continuously differentiable with respect to S) so that we can apply Itô's formula. Let's model the asset price S as a geometric Brownian motion satisfying under the risk-neutral measure

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (3.30)$$

Then by Ito's lemma and taking expectations of both sides we have

$$\mathbb{E}[v(t + \delta, S_{t+h})] = v(t, S) + \mathbb{E}_t \left[\int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, S_u) + \frac{\partial v}{\partial S}(u, S_u) r S_u + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(u, S_u) \sigma^2 S_u^2 \right) du \right] \quad (3.31)$$

Now let's substitute (3.29) into the LHS, which gives us

$$e^{rh} v(t, S) \geq v(t, S) + \mathbb{E}_t \left[\int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, S_u) + \frac{\partial v}{\partial S}(u, S_u) r S_u + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(u, S_u) \sigma^2 S_u^2 \right) du \right] \quad (3.32)$$

We can use the Taylor approximation $e^{rh} = 1 + rh + o(h)$, which permits us to cancel the $v(t, S)$ on both sides. Dividing both sides by h in the same step, we get

$$rv(t, S) + \frac{o(h)}{h} \geq \mathbb{E}_t \left[\frac{1}{h} \int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, S_u) + \frac{\partial v}{\partial S}(u, S_u) r S_u + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(u, S_u) \sigma^2 S_u^2 \right) du \right] \quad (3.33)$$

Sending $h \rightarrow 0$ makes the higher order term $o(h)/h$ vanish, and the expectation of the integral to collapse to the integrand. Thus we obtain

$$rv(t, S) \geq \frac{\partial v}{\partial t}(t, S) + \frac{\partial v}{\partial S}(t, S) r S + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(t, S) \sigma^2 S^2 \quad (3.34)$$

or for simplicity, we define the linear parabolic operator $\mathcal{L}$ [8]:

$$\mathcal{L}[v] = rv(t, S) - \frac{\partial v}{\partial t}(t, S) r S - \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(t, S) \sigma^2 S^2 \quad (3.35)$$

Thus we can concisely express this inequality as

$$\min \left(-\frac{\partial v}{\partial t}(t, S) + \mathcal{L}[v], v(t, S) - g(S) \right) \geq 0 \quad (3.36)$$

However, note that at time t , we must either exercise the option or do nothing and continue. One of these two options must be optimal, implying that one of the arguments must be 0. Thus we actually have an equality:

$$\min \left(-\frac{\partial v}{\partial t}(t, S) + \mathcal{L}[v], v(t, S) - g(S) \right) = 0 \quad (3.37)$$

which is the variational equation [18] for the American option. Note this is exactly the formulation of the PINN problem (2.36).

We are now ready to derive the boundary conditions. Our option has a terminal payoff at time T given by $g(S)$, so we have the terminal condition

$$v(T, S) = g(S) \quad (3.38)$$

which satisfies the form of the terminal condition for PINNs (2.37). For the spatial boundary conditions, we have two boundaries to consider: $S \rightarrow 0$ and $S \rightarrow \infty$. In practice, we will have to truncate the domain to $[S_{\min}, S_{\max}]$ for some sufficiently small $S_{\min}$ and sufficiently large $S_{\max}$. At the lower boundary, we have the boundary condition

$$v(t, S_{\min}) = g(S_{\min}) \quad (3.39)$$

because it is always optimal to exercise the option when the asset is at its lowest price. At the upper boundary, we assume the option is too far out of the money to ever finish in the money, and hence the option price is approximately 0. Thus we have the boundary condition

$$v(t, S_{\max}) = 0 \quad (3.40)$$

These three conditions together would form the boundary setup (2.38) for our PINN.

3.3.2 Reduction of multi-asset to one-asset case

Before we apply the PINN framework to the multi-dimensional case, we will first introduce a benchmark for the multi-dimensional case, from which we will build the boundary conditions for the PINN, as they change depending on the payoff function. The benchmark will need to be able to be reduced to a one-dimensional problem, so that we can solve it using the binomial tree method as a reference for our PINN results. This will, in turn, give us confidence that our PINN implementation is correct, and that the results we get for more complex payoffs are reasonable.

In the multi-dimensional case, we have k underlying assets with price processes following a correlated geometric Brownian motion. Mathematically, let each asset S^i satisfy under the risk-free measure $\mathbb{Q}$, for $i = 1, \dots, k$,

$$dS_t^i = rS_t^i dt + \sigma_i S_t^i dW_t^i \quad (3.41)$$

with $\langle W^i, W^j \rangle_t := \mathbb{E}[dW_t^i dW_t^j] = \rho_{ij} dt$ for all i, j pairs. As the geometric Brownian motion is multiplicative, for our benchmark, we will consider the put on the product of the assets. More precisely, we consider the put option with payoff

$$g(S^1, \dots, S^k) = \left(K - \prod_{i=1}^k S^i \right)^+ \quad (3.42)$$

To represent the dynamics of the product of the assets, we can apply Ito's formula to the log of the price process. The log representation is

$$\ln S_t^i = \ln S_0^i + \int_0^t \left(r - \frac{1}{2} \sigma_i^2 \right) ds + \int_0^t \sigma_i dW_s^i \quad (3.43)$$

Define our new asset X to be

$$X_t := \prod_{i=1}^k S_t^i \quad (3.44)$$

and hence we have

$$\ln X_t = \sum_{i=1}^k \ln S_t^i = \ln X_0 + \int_0^t \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) ds + \sum_{i=1}^k \int_0^t \sigma_i dW_s^i \quad (3.45)$$

By taking the differential we have

$$d \ln X_t = \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) dt + \sum_{i=1}^k \sigma_i dW_t^i \quad (3.46)$$

The quadratic variation of the martingale term is

$$\left\langle \sum_{i=1}^k \sigma_i W^i, \sum_{j=1}^k \sigma_j W^j \right\rangle_t = \sum_{i=1}^k \sum_{j=1}^k \sigma_i \sigma_j \rho_{ij} t =: \sigma_X^2 t \quad (3.47)$$

i.e. $\sigma_X^2 = \sum_{i=1}^k \sum_{j=1}^k \sigma_i \sigma_j \rho_{ij}$. So we can express the second term in terms of a new Brownian motion $\widetilde{W}$:

$$\sum_{i=1}^k \sigma_i dW_s^i = d\widetilde{W}_t \quad (3.48)$$

Going back, we have

$$d \ln X_t = \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) dt + \sigma_X d\widetilde{W}_t \quad (3.49)$$

Exponentiating using Ito's formula gets us

$$dX_t = X_t \left[\left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) dt + \sigma_X d\widetilde{W}_t \right] + \frac{1}{2} X_t \sigma_X^2 dt \quad (3.50)$$

which we can rearrange to get

$$dX_t = \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 + \frac{1}{2} \sigma_X^2 \right) X_t dt + \sigma_X X_t d\widetilde{W}_t \quad (3.51)$$

which is a geometric Brownian motion with volatility σ_X and drift r_X where

$$r_X = kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 + \frac{1}{2} \sigma_X^2 \quad (3.52)$$

$$= kr + \sum_{i < j} \sigma_i \sigma_j \rho_{ij} \quad (3.53)$$

by matching coefficients. We can verify the equivalence by simulating many price paths for the assets and comparing their product against the simulations of price paths for the equivalent asset. Using the initial values

$$r = 0.05 \quad S_0 = (10, 5, 3)^\top \quad \sigma = (0.2, 0.3, 0.1)^\top \quad R = \begin{pmatrix} 1 & 0.6 & -0.2 \\ 0.6 & 1 & 0.4 \\ -0.2 & 0.4 & 1 \end{pmatrix} \quad (3.54)$$

where R is the correlation matrix, we can plot the histogram of final prices in Figure [3.1].

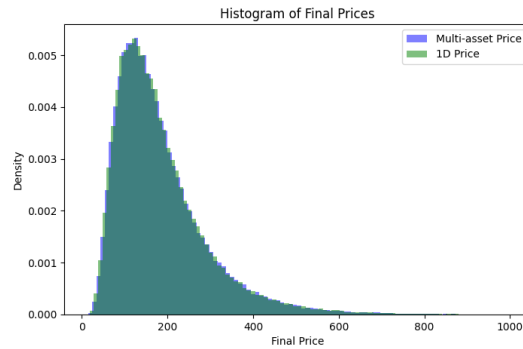


Figure 3.1 Histogram of final prices for product of assets and equivalent asset

They look very similar, and we can further statistically test if they come from the same distribution. Using Kolmogorov-Smirnov's 2-sample test on the null hypothesis that they come from the same distribution, we obtain a p-value of 0.8491, which does not provide sufficient evidence to reject the null hypothesis, which gives us stronger evidence that our derivation is correct.

Therefore, we will be able to price the put option on the product of multiple assets and compare that to the price of the put option on this single constructed asset as a benchmark.

3.3.3 PINN formulation for multi-dimensional case

With the benchmark in place, we now need to derive the variational inequality for the multi-dimensional case where we have k assets. This is a straightforward extension of the one-dimensional case, where the value function $v(t, S^1, \dots, S^k)$ now takes $k+1$ inputs. For notation, we will express the multiple assets as a vector $\mathbf{S} = (S^1, \dots, S^k)^\top$, so the value function is $v(t, \mathbf{S})$. Similarly, the payoff function will be written as $g(S^1, \dots, S^k) = g(\mathbf{S})$.

The value function stays

$$v(t, \mathbf{S}) = \sup_{\tau \in [t, T]} \mathbb{E}_t [e^{-r(\tau-t)} g(\mathbf{S}_\tau) \mid \mathbf{S}_t = \mathbf{S}] \quad (3.55)$$

Should we choose not to exercise the option at time t , we can apply the same argument as in the one-dimensional case: considering a small $h > 0$, we can apply Itô's lemma on multiple dimensions to yield

$$\begin{aligned} v(t+h, \mathbf{S}_{t+h}) = v(t, \mathbf{S}) &+ \int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, \mathbf{S}_u) + \sum_{i=1}^k \frac{\partial v}{\partial S^i}(u, \mathbf{S}_u) r S_u^i \right. \\ &\left. + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(u, \mathbf{S}_u) \sigma_i \sigma_j \rho_{ij} S_u^i S_u^j \right) du \end{aligned} \quad (3.56)$$

The steps from here are identical to the one-dimensional case, where we can take expectations of both sides and use the second order Taylor expansion of the exponential to get the variational inequality

$$rv(t, \mathbf{S}) \geq \frac{\partial v}{\partial t}(t, \mathbf{S}) + \sum_{i=1}^k \frac{\partial v}{\partial S^i}(t, \mathbf{S}) r S^i + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(t, \mathbf{S}) \sigma_i \sigma_j \rho_{ij} S^i S^j \quad (3.57)$$

where to simplify into the PINN framework as in (2.36), we can define the operator $\mathcal{L}$ as

$$\mathcal{L}[v] = rv(t, \mathbf{S}) - \sum_{i=1}^k \frac{\partial v}{\partial S^i}(t, \mathbf{S}) r S^i - \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(t, \mathbf{S}) \sigma_i \sigma_j \rho_{ij} S^i S^j \quad (3.58)$$

and the problem reduces to the same variational inequality as in the one-dimensional case (3.37), with a different operator $\mathcal{L}$.

To finish the PINN formulation, we need to derive the boundary conditions. The terminal condition is straightforward, as it is given by the payoff function:

$$v(T, \mathbf{S}) = g(\mathbf{S}) = \left(K - \prod_{i=1}^k S^i \right)^+ \quad (3.59)$$

For the spatial boundary conditions, we will again truncate the domain to

$$\Omega = \left\{ \mathbf{S} = (S^1, \dots, S^k) \in \mathbb{R}^k : S^i \in [S_{\min}^i, S_{\max}^i] \text{ for } i = 1, \dots, k \right\} \quad (3.60)$$

for some sufficiently small and large respective minimum and maximum values for each stock. The lower boundary will be sampled from the set

$$\partial_{\text{lower}} \Omega = \left\{ \mathbf{S} \in \Omega : S^i = S_{\min}^i \text{ for some } i \right\} \quad (3.61)$$

where we have the boundary condition

$$v(t, \mathbf{S}) = g(\mathbf{S}), \quad \mathbf{S} \in \partial_{\text{lower}}\Omega \quad (3.62)$$

and the upper boundary will be sampled from the set

$$\partial_{\text{upper}}\Omega = \{\mathbf{S} \in \Omega : S^i = S_{\text{max}}^i \text{ for some } i\} \quad (3.63)$$

with the boundary condition

$$v(t, \mathbf{S}) = 0, \quad \mathbf{S} \in \partial_{\text{upper}}\Omega \quad (3.64)$$

analogous to the one-dimensional case. Of course, this framework could be easily adapted to other payoff functions, where the boundary conditions would change accordingly (but not the variational inequality). But for our benchmark, this is the setup we will use.

3.4 Sobolev Deep Neural Networks

For the Black-Scholes model, we will additionally explore a different method to price American puts with neural networks, but this time instead of the traditional boundary conditions, we will work with fractional Sobolev norms to construct our loss function. This is because it has been shown that conditional on the loss converging to 0, the neural network will converge to the true solution [8], which is a stronger guarantee than just pointwise convergence. Additionally, the loss function components no longer depend on the payoff and is applicable for all situations, a desirable property for a general method.

The theory of Sobolev spaces is rich and we will not attempt to cover all the details, to avoid diverting attention from the main focus of the project. However, we make particular use of fractional Sobolev spaces, which are a generalisation of the integer order Sobolev spaces.

3.4.1 Sobolev spaces

Throughout this section, let $\Omega \subset \mathbb{R}^n$ be an open bounded set with Lipschitz boundary, on which our target function is defined, and let $p \in [1, +\infty)$ be the integrability parameter. Recall the Lebesgue space

$$L^p(\Omega) := \left\{ u : \Omega \rightarrow \mathbb{R} : u \text{ is measurable and } \int_{\Omega} |u(x)|^p dx < \infty \right\} \quad (3.65)$$

and the integer order Sobolev space $W^{m,p}(\Omega)$ for $m \in \mathbb{N}$ [19],

$$W^{m,p}(\Omega) := \{u \in L^p(\Omega) : D^{\alpha}u \in L^p(\Omega), |\alpha| \leq m\}, \quad (3.66)$$

where $D^{\alpha}u$ denotes the weak derivative of u with respect to the multi-index α .

The fractional Sobolev space $W^{s,p}(\Omega)$ is defined with the fractional exponent $s \in (0, 1)$ as [20]:

$$W^{s,p}(\Omega) := \left\{ u \in L^p(\Omega) : \frac{|u(x) - u(y)|}{|x - y|^{\frac{n}{p} + s}} \in L^p(\Omega \times \Omega) \right\}, \quad (3.67)$$

which can be interpreted as an intermediate Banach space between $L^p(\Omega)$ and $W^{1,p}(\Omega)$, endowed with the norm

$$\|u\|_{W^{s,p}(\Omega)}^p = \|u\|_{L^p(\Omega)}^p + [u]_{W^{s,p}(\Omega)}^p \quad (3.68)$$

$$= \int_{\Omega} |u|^p dx + \int_{\Omega} \int_{\Omega} \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} dx dy, \quad (3.69)$$

where we have used the *Gagliardo seminorm* of u :

$$[u]_{W^{s,p}(\Omega)} := \left(\int_{\Omega} \int_{\Omega} \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} dx dy \right)^{\frac{1}{p}}. \quad (3.70)$$

We can extend this definition to non-integer $s > 1$. In the decomposition $s = m + \sigma$ with $m \in \mathbb{N}$ and $\sigma \in (0, 1)$, we define

$$W^{s,p}(\Omega) := \{u \in W^{m,p}(\Omega) : D^{\alpha}u \in W^{\sigma,p}(\Omega), |\alpha| = m\} \quad (3.71)$$

endowed with the natural norm

$$\|u\|_{W^{s,p}(\Omega)}^p := \|u\|_{W^{m,p}(\Omega)}^p + \sum_{|\alpha|=m} [D^{\alpha}u]_{W^{\sigma,p}(\Omega)}^p. \quad (3.72)$$

In the special case $p = 2$, the fractional Sobolev space becomes a Hilbert space, which we denote $H^s(\Omega) := W^{s,2}(\Omega)$. This is the specific case we will be working with in our loss construction.

3.4.2 Model

Suppose we would like to price an option on k assets $\mathbf{S} = (S^1, \dots, S^k)$. As asset prices are theoretically unbounded, we truncate it in our model. Thus the domain of the target function can be chosen to be the open set

$$\Omega = (S_{\min}^1, S_{\max}^1) \times \dots \times (S_{\min}^k, S_{\max}^k) \quad (3.73)$$

In our study we will make the simplification that $S_{\min}^i = S_{\min}^j$ and $S_{\max}^i = S_{\max}^j$ for all $i \neq j$, which we can call a and b respectively. Our spaces would consequently have the definitions

$$\Omega = (a, b)^k \quad \Omega_T = (0, T) \times \Omega \quad \Sigma = (0, T) \times \partial\Omega \quad (3.74)$$

where $\partial\Omega =: \Gamma$ is the boundary of Ω , a hyper-rectangle. We will use the representation

$$\Gamma = \bigcup_{i=1}^k (\{x \in [a, b]^k : x_i = a\} \cup \{x \in [a, b]^k : x_i = b\}) \quad (3.75)$$

which is not the simplest, but it is useful as it is decomposed into faces, which is useful when we would like to use the norm derivative later. Note that in the one-dimensional case, this reduces simply to the points $\Gamma = \{a, b\}$.

3.4.3 Loss function

Our loss function would be a weighted sum of four loss components $\sum_{i=1}^4 \lambda_i J_i$, each of which ensures regulation of a specific aspect of our function. We will use $k = 2$ in our experiments, but the loss construction is applicable for any k .

1. The variational loss, which is as before, controlling the accuracy of our function in the interior of our domain:

$$J_1 = \|\min\{\mathcal{G}[f], f - g\}\|_{L^2(\Omega_T)}^2 \quad (3.76)$$

2. The energy loss: before we were only controlling the L2 norm of the payoff, but we require the option value to equal the payoff as a function and not just pointwise. Thus we need to incorporate the gradient into the loss so that the spatial slope of the payoff is respected.

$$J_2 = \|f(0, \cdot) - g(\cdot)\|_{H^1(\Omega)}^2 \quad (3.77)$$

where we have the norm defined as

$$\|d\|_{H^1(\Omega), \mu_2}^2 = \|d\|_{L^2(\Omega)}^2 + \|\nabla d\|_{L^2(\Omega)}^2 \quad (3.78)$$

and the derivative

$$\nabla d = (\partial_{x_1} d, \dots, \partial_{x_k} d) \quad (3.79)$$

3. The Dirichlet trace loss:

$$J_3 = \|f(t, x) - g(x)\|_{H^{\frac{3}{4}, \frac{3}{2}}(\Sigma)}^2 \quad (3.80)$$

where

$$\|d\|_{H^{\frac{3}{4}, \frac{3}{2}}(\Sigma)}^2 = \|d\|_{H^{0,1}(\Sigma)}^2 + \int_0^T \int_0^T \frac{\|d(t_1, \cdot) - d(t_2, \cdot)\|_{L^2(\Gamma)}^2}{|t_1 - t_2|^{2.5}} dt_1 dt_2 \quad (3.81)$$

$$+ \int_0^T \int_{\Gamma} \int_{\Gamma} \frac{|d(t, x) - d(t, y)|^2}{\|x - y\|^k} d\sigma(x) d\sigma(y) dt \quad (3.82)$$

The time exponent 2.5 is equal to $2p + 1$ where p is the decimal part of the time fractional exponent, and the space exponent is equal to $2\theta + k - 1$ where θ is the decimal part of the space fractional exponent, and k the dimension (so would be 2 in two dimensions). Note that here $d\sigma(x)$ and $d\sigma(y)$ indicate the surface measures on Γ , but in practice we can treat these as dx and dy .

4. The Neumann trace loss:

$$J_4 = \|\nabla(f - g) \cdot \nu\|_{H^{\frac{1}{4}, \frac{1}{2}}(\Sigma)}^2 \quad (3.83)$$

where the norm derivative is $\nabla(f - g) \cdot \nu = \partial_{x_{i^*}}(f - g)$ given that $x_{i^*} \in \{a, b\}$ and $x_i \in (a, b)$ for $i \neq i^*$. Rigorously, we require the sign but in practice this is not necessary, as we will square in the end. The fractional norm is thus written

$$\|d\|_{H^{\frac{1}{4}, \frac{1}{2}}(\Sigma)}^2 = \|d\|_{L^2(\Sigma)}^2 + \int_0^T \int_0^T \frac{\|d(t_1, \cdot) - d(t_2, \cdot)\|_{L^2(\Gamma)}^2}{|t_1 - t_2|^{1.5}} dt_1 dt_2 \quad (3.84)$$

$$+ \int_0^T \int_{\Gamma} \int_{\Gamma} \frac{|d(t, x) - d(t, y)|^2}{\|x - y\|^k} d\sigma(x) d\sigma(y) dt \quad (3.85)$$

because $H^{0,0} = L^2$.

The third and fourth losses are those that enforce f to solve the option pricing variational inequality with regularity in the time and space dimensions beyond pointwise [21]. In practice, we will compute a Monte Carlo estimate of these integrals by sampling pairs (t_1, t_2) from the domain, evaluating the integrand and taking the mean across all pairs.

3.5 Numerical Results

3.5.1 Tree benchmark against closed-form solution

First we will evaluate the performance of the benchmark binomial tree method for the one-dimensional case against the closed-form solution, which we would take as the ground truth.

To do this, we will compute the tree option price without taking the maximum of the payoff and continuation value at each step so that we can compare the tree price against the Black-Scholes price.

First, we will evaluate the tree's accuracy on the most sensitive point, which is at time $t = 0$ and when the initial asset price S is equal to the strike price K (at the money). Figure 3.2 shows the convergence of the tree price to the closed-form Black-Scholes price for European put options as we increase the depth of the tree. We can see that around 1000 steps yields a sufficiently close price to the closed-form solution, and further increasing the depth does not yield significant improvements, so this is the depth we will use for our tree benchmark.

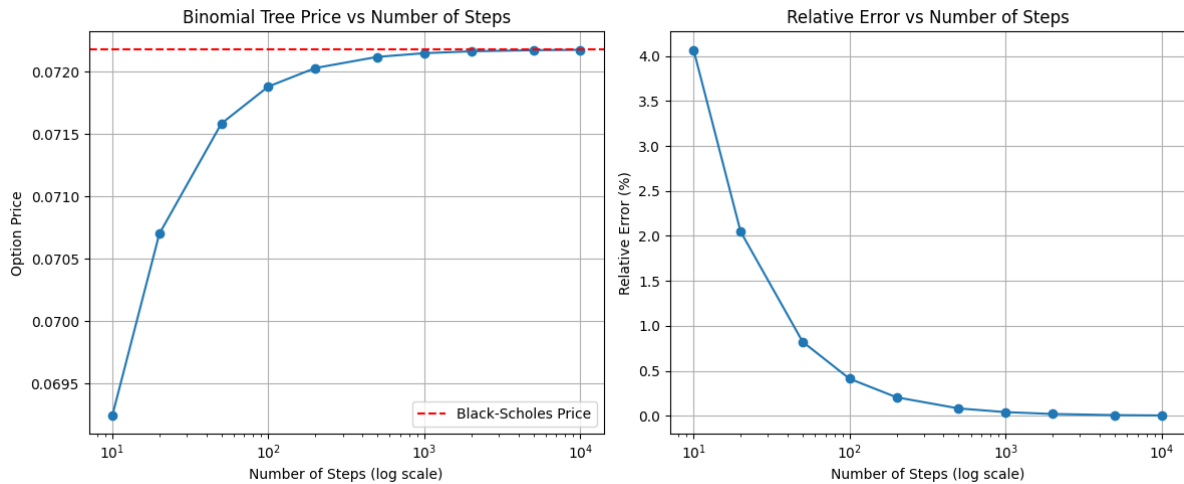


Figure 3.2 Tree convergence at $t = 0, S = K$ over different depths

Evaluating the tree price against the closed-form solution for different initial asset prices S at the initial time $t = 0$ in Figure 3.3, we can see that the tree price is very close to the ground truth, with less than 0.05% of relative error until we are far out of the money where the option price is very close to 0 and unsurprisingly yields greater errors, where machine precision becomes a limiting factor.

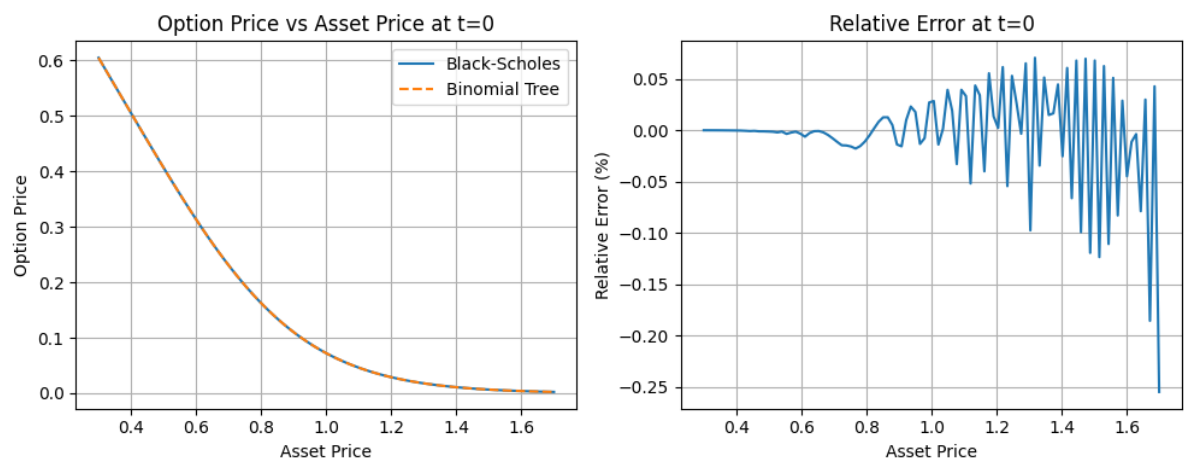


Figure 3.3 Comparison of tree and closed-form prices for European put option with varying initial asset price

We can see that the tree price also closely tracks the term structure of the option price, as shown in Figure 3.4, where the error is again less than 0.05%, decreasing as we get closer to maturity,

which is expected as the tree has not done as many backward-induction steps to get to the initial time, where errors can accumulate.

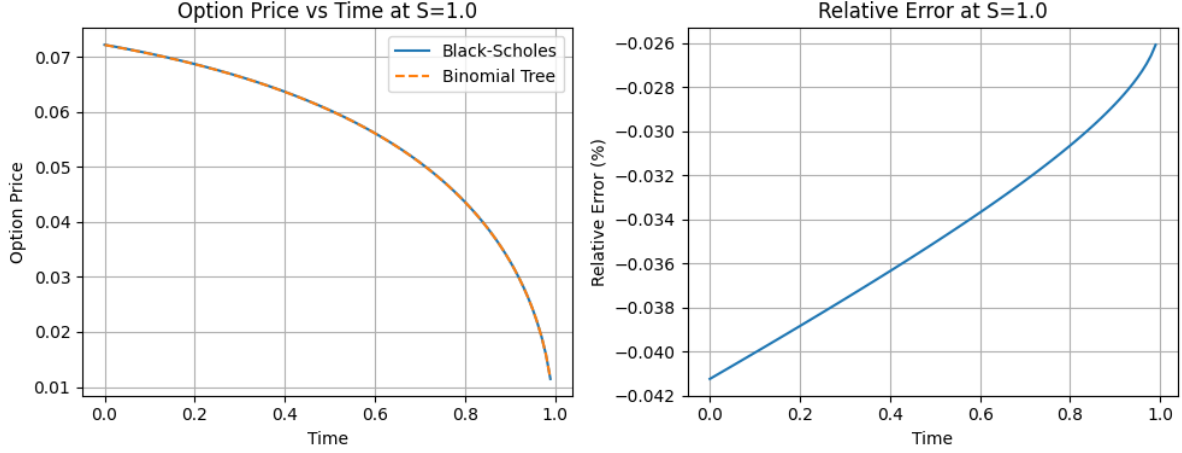


Figure 3.4 Comparison of tree and closed-form prices for European put option with varying time

Overall, we can conclude that the tree benchmark yields very strong results against the closed-form solution, giving us good confidence that it will provide accurate results for American options too, where we do not have a closed-form solution to compare against. This will serve as our benchmark for evaluating the performance of our PINN implementation for American options.

3.5.2 PINN results against tree benchmark

We will now train our PINN on the one-dimensional case and compare the results against the tree benchmark on the American put on the asset with parameters $K = 1, r = 0.1, \sigma = 0.3, T = 1$ across a range of possible initial asset prices $S_0 \in [0, 3]$. This will be done through a feedforward neural network with 3 hidden layers each of width 32, with learning rate 0.001, batch size 10000 with 30000 maximum epochs with early stopping patience of 500 epochs. We train 25 models with different random initialisation and compute the ensemble mean for our final price estimate.

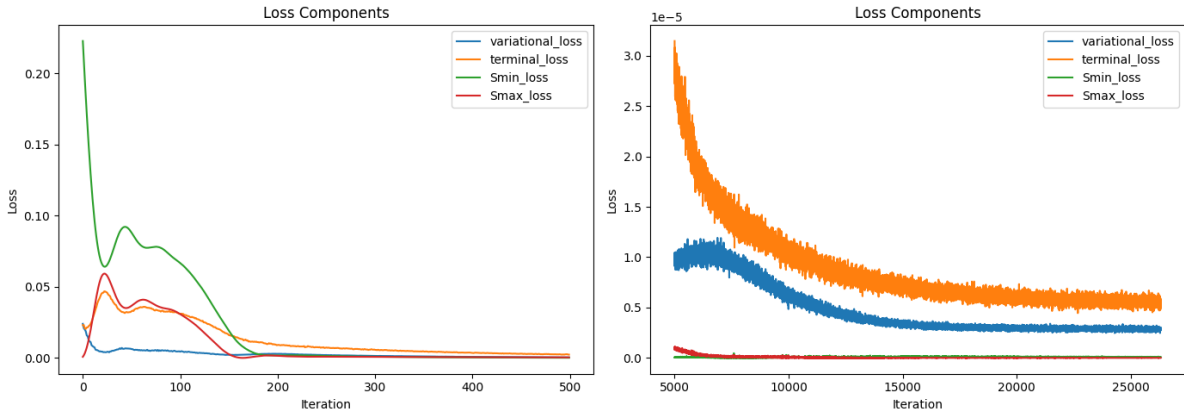


Figure 3.5 Loss convergence for the one-dimensional PINN for one run

Figure 3.5 shows the convergence of the loss function for one run, and we can see that the loss

converges well, with each of them having similar magnitudes, converging in the order of 10^{-5} .

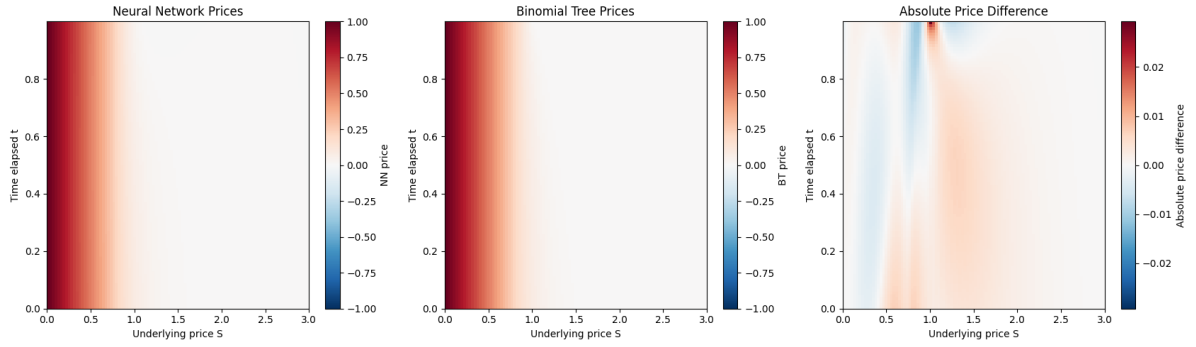


Figure 3.6 PINN vs tree price comparison heatmap across domain

Figure 3.6 shows the heatmap of the PINN price against the tree price across the domain of time and initial asset price. We can see that the PINN price is very close to the tree price across the domain, where the time value is correctly captured at the most sensitive point of $S = K$ and $t = 0$ as seen in Figure 3.7. This figure additionally shows the model capturing the term structure of the option price, where the PINN price closely tracks the tree price across time for $S = K$.

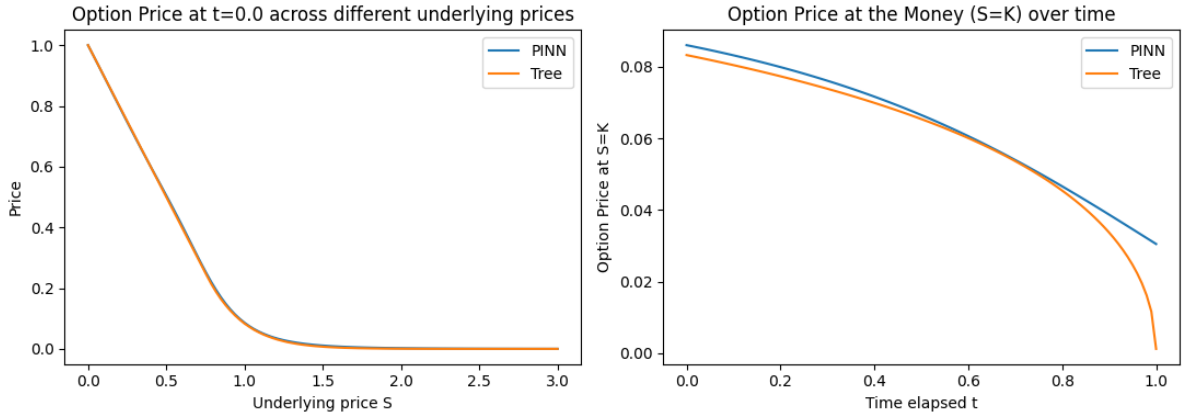


Figure 3.7 Price comparison for varying underlying price, and for term structure at $S = K$

We can also zoom in on the most sensitive point of $(t, S) = (0, K)$ and evaluate the error. Table 3.1 shows the mean PINN price across 30 runs with its 95% confidence interval, and we can see that the relative error is around 3%, which is good considering we were sampling across the whole domain and not just at this point. However, we can see that the benchmark tree price lies around 4 standard deviations away from the PINN mean price, suggesting that there is some bias in the PINN approximation.

Tree price	PINN price	Relative error
0.0833	0.0860 ± 0.0006	3.33%

Table 3.1 PINN price at the money and initial time against tree benchmark

We can investigate this apparent bias by tracking the price $f(t = 0, S = K)$ across different training epochs, as shown in Figure 3.8. We can see that the price spikes above the tree price and then settles around the marked benchmark tree price but interestingly approaches it from below, confirming our suspicion that the PINN price is biased against the tree price. That said, the tree price is not guaranteed to be the ground truth.

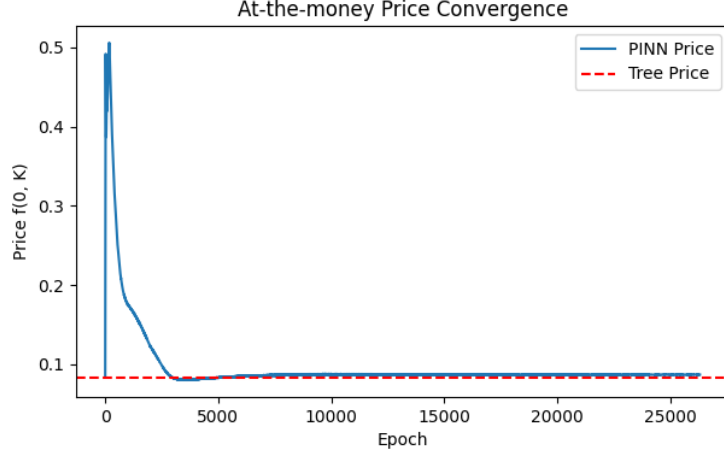


Figure 3.8 $f(0, K)$ across training epochs for one run

3.5.3 Sobolev PINN results

We will now train a PINN with the Sobolev loss function, with the same model architecture and training setup as the vanilla PINN, and compare the results against the tree benchmark. We can see from Figure 3.9 that the Sobolev PINN price also gets close to the tree price across the domain, but the inaccuracy here is more significant than the vanilla PINN, especially at the $(t, S) = (0, K)$ point, which is arguably the most important point. However, considering that the loss function isn't manufactured from the payoff function, we can see that this method could be effective for more general problems where boundary conditions are not as straightforward to implement, and the price is still in the right ballpark.

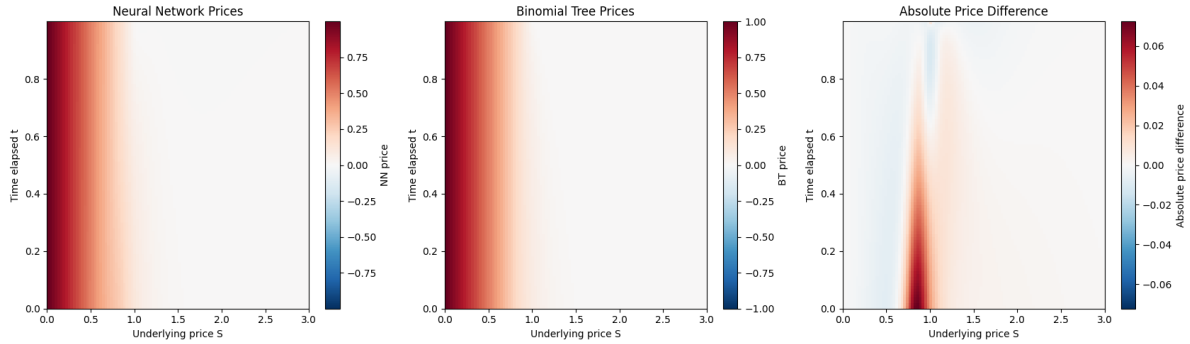


Figure 3.9 Sobolev PINN vs tree price comparison heatmap across domain

We can see especially in Figure 3.10 that the Sobolev PINN manages to capture the term structure of the option price with the theta collapse as we approach maturity, and the error is more significant due to a hump in the price against asset price around the strike price. This suggests that the model is capable of a good fit across the domain, but may be undertrained or have too small a capacity.

Overall this method is promising, but it may require more careful training and tuning to get better results as presented in [8], who demonstrated its effectiveness through extensive training on the GPU with much larger networks, which is beyond my scope. With the one-dimensional case already underperforming the vanilla PINN, it is unsurprising that the multi-dimensional case yielded me less accurate results, so for the multi-dimensional case and studies on the free boundary, I will only present the results for the vanilla PINN, which is already a good method

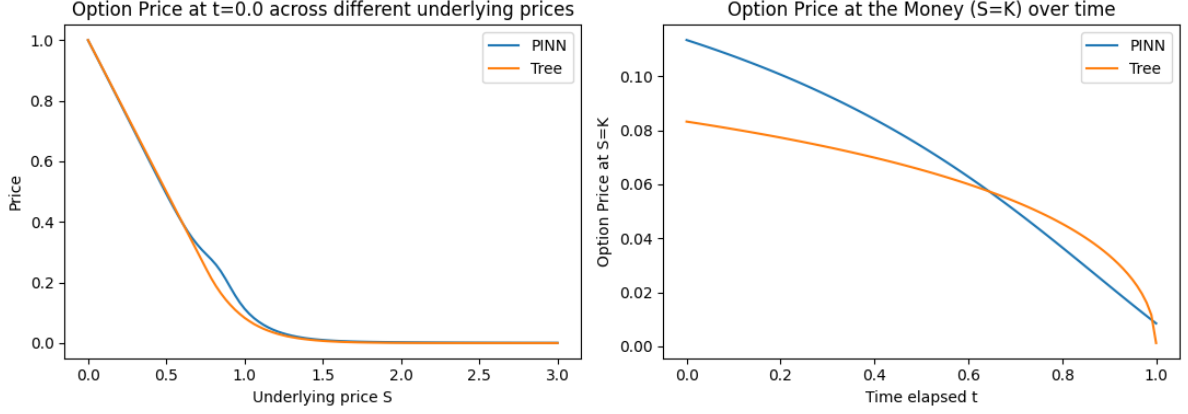


Figure 3.10 Price comparison for varying underlying price, and for term structure at $S = K$ with the Sobolev PINN

for this problem. That said, it was interesting nonetheless to see how we could construct the loss function from another angle altogether, without relying on manual boundary conditions.

3.5.4 Multi-dimensional PINN

We move onto training a PINN on multiple assets, for simplicity and visualisation purposes, we will train a two-dimensional PINN on the put option on the product of two assets, with parameters $K = 1, r = 0.1, T = 1$ with stock volatilities $\sigma = (0.3, 0.2)^\top$ and correlation between assets $\rho = 0.5$. Both of these assets will be truncated to $[0, 3]$ and we will sample uniformly from the whole domain for training. The model architecture will be similar to the one-dimensional case, with a bigger feedforward neural network of 4 hidden layers each of width 64, learning rate 0.001 and batch size 10000. This will be trained for a maximum of 40000 epochs with early stopping patience of 500 epochs across 10 different seeds. Figure 3.11 shows the heatmap of the PINN price against the tree price, where we take the S_1 - S_2 cross section and fix time at $t = 0$. We can see that the PINN price is close to the tree price across the domain, with some stronger bias at the in-the-money regions where the price is higher and hence is not as significant when we take relative error into consideration.

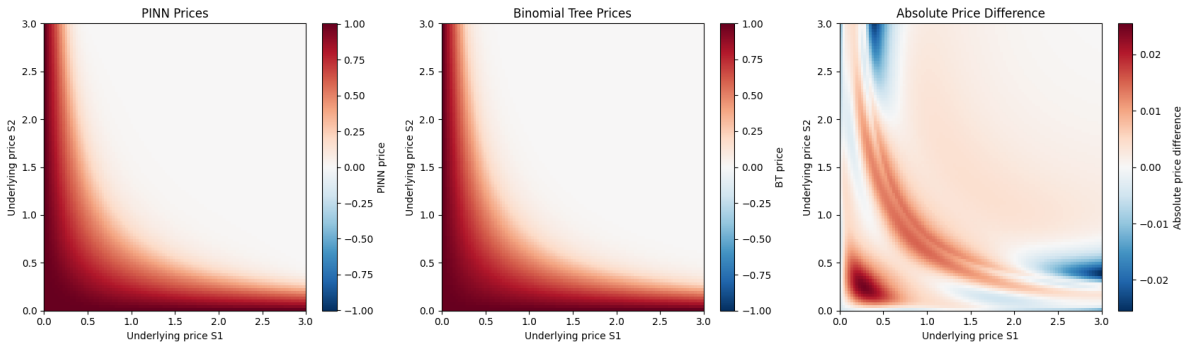


Figure 3.11 PINN vs tree price comparison heatmap across S_1 - S_2 cross section at $t = 0$

We can zoom in on the $S_1 \times S_2 = K$ cross section that we can see clearly in the heatmap, and plot the price along this cross section across time, shown in the left figure of Figure 3.7. We can see that the PINN fit is satisfactory, with inconsistencies at the tails due to our boundary conditions, which are set at $S = 0$ and $S = 3$ for both assets, which are not the true boundaries

of the problem and a model simplification we made.

Tree price	PINN price	Relative error
0.0993	0.1060 ± 0.0048	6.75%

Table 3.2 Prices $f(0, \sqrt{K}, \sqrt{K})$ for the two-dimensional PINN against tree benchmark

Overall, by Table 3.2, we can see that the relative error at the most sensitive point at the triple $(0, \sqrt{K}, \sqrt{K})$ is around 7% which is considerably higher than the one-dimensional case, which is expected as we are now sampling across a much larger domain, and we are restrained by model capacity and training time due to personal hardware constraints, which is a common issue for multi-dimensional PINNs.

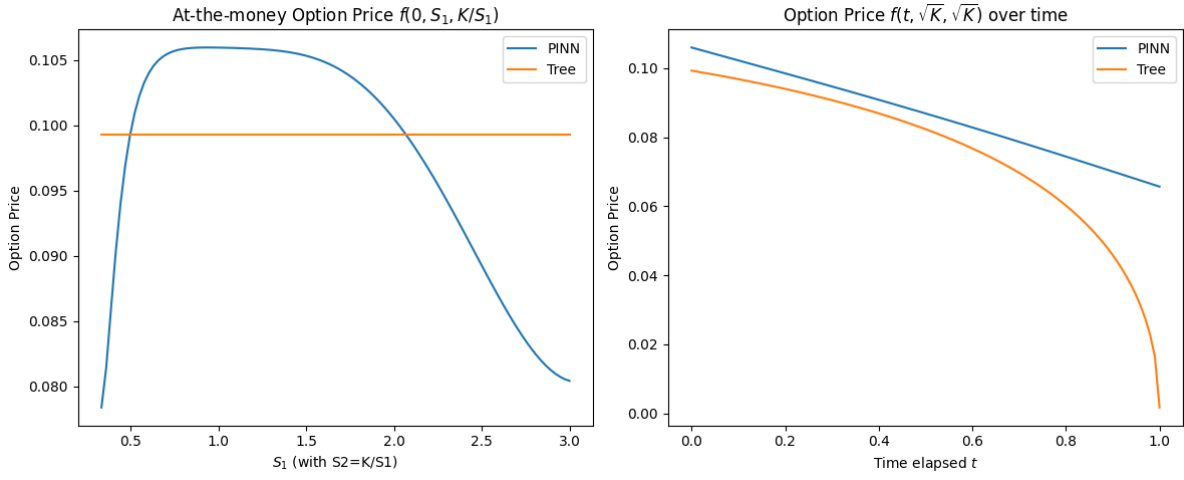


Figure 3.12 PINN vs tree price comparison for at the money cross section and term structure

We can also take the cross section at $S_1 = S_2$ to evaluate the term structure of the option price, as shown in the right figure of Figure 3.12. We can see that the PINN has managed to capture the theta decay of the option price, but the consistent bias of the price is evident here and the model struggles to capture the more rapid theta decay as we approach maturity. This may be due to the fact gamma approaches a dirac delta function at the money as we approach maturity, whose sharp feature is difficult for the model to capture when our samples are sparser in the expanded domain.

3.5.5 Free boundary

One-dimensional case

For the one-dimensional free boundary, we will first evaluate the free boundary from the tree benchmark. Here, the continuation values are extracted through a small modification to the tree algorithm, where we compute the price of the Bermudan option with the same parameters with possibility of exercise at every time step except for at $t = 0$, effectively serving as a continuation value of the American option at $t = 0$. This continuation value is compared with the intrinsic value (the payoff) like we introduced in equation (2.9) in the mathematical background. The left figure in Figure 3.13 shows the continuation probabilities across different (t, S) pairs, and here we use hard classification as binomial tree pricing is deterministic.

We can see that the free boundary is correctly captured at glance, where we exercise the option when the asset price is sufficiently low, whose threshold increases as we get closer to maturity,

consistent with the intuition that the option is losing time value.

We now evaluate the free boundary from the PINN. The central figure in Figure 3.13 shows the continuation probabilities across different (t, S) pairs using the sigmoid distribution as described in equation (2.12). We can see that the shape of the free boundary is similar to the tree benchmark, and with soft classification the lowest probability is around 0.4, which is not unintuitive as in the exercise region, the continuation value is theoretically equal to the payoff, and hence we should be indifferent between exercising and continuing.

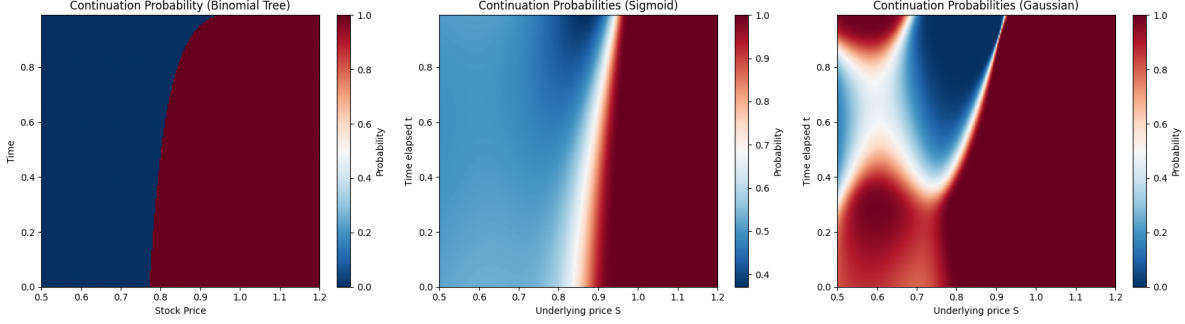


Figure 3.13 Estimated continuation probabilities from PINN for the American put option

To use the Gaussian method, we will first evaluate its validity by plotting the distribution of the standardised price estimates across all seeds in Figure 3.14, noting that as the variance is unknown and we use the sample variance, the standardised price estimates would follow a t-distribution with $n - 1$ degrees of freedom where n is the number of seeds.

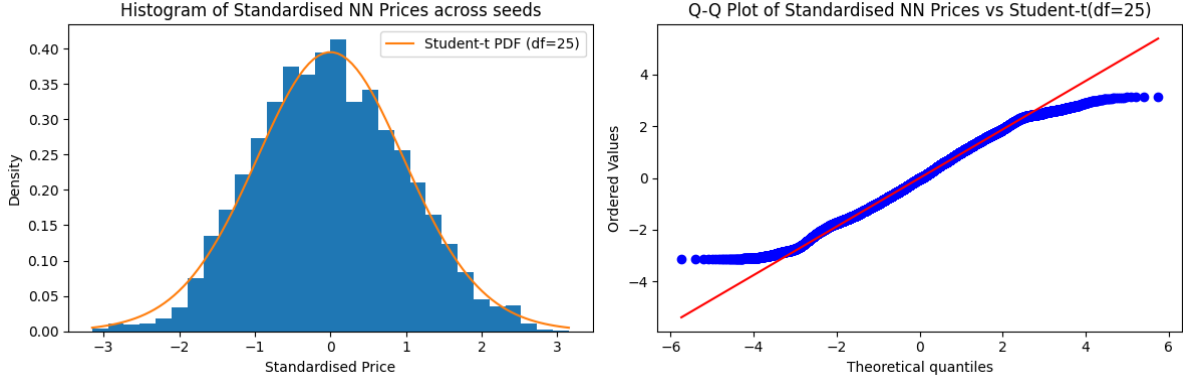


Figure 3.14 Distribution of standardised neural network price estimates across 25 runs

By looking at the histogram and the Q-Q plot, we can see that the distribution of the standardised price estimates is reasonably close to a t-distribution, with the exception of the tails. We see that the observed distribution has lighter tails than the theoretical t-distribution, which is more statistically acceptable, so we will nevertheless proceed with caution in evaluating the free boundary using the Gaussian method. The right figure in Figure 3.13 shows the continuation probabilities across different (t, S) pairs.

We can see that the shape roughly corresponds to the free boundaries we have seen before, but this is a much noisier estimate, which puts into question the assumptions we have made. We are essentially assuming that the price estimates of the PINN are unbiased and normally distributed around the true price, which is quite strong and not true in practice, as we have already observed in Figure 3.6 where we have zones where the PINN price is systematically above or below the tree price, suggesting bias.

Two-dimensional case

Using the same methodology as the one-dimensional case, we plot our estimated free boundaries in Figure 3.15. This is done on the S_1 - S_2 cross section at $t = 0$, with the additional reference line of $S_1 \times S_2 = K$, which should be a continuation region as it is at the money and is worthless under immediate exercise.

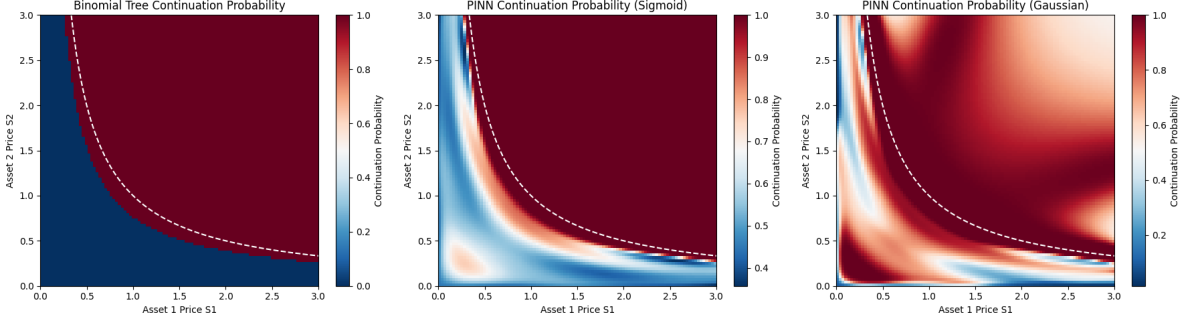


Figure 3.15 Estimated continuation probabilities for 2-dimensional case with $S_1 \times S_2 = K$ reference line

We can see that both methods of estimating the continuation probabilities yield similar shapes to the binomial tree benchmark, with the Gaussian method being noisier than the sigmoid method. We can further examine the continuation probabilities over time along the $S_1 = S_2$ cross section, varying across time this time. This is shown in Figure 3.16, where we can see that the PINN has successfully captured the structure of the free boundary across time where $S_1 = S_2$.

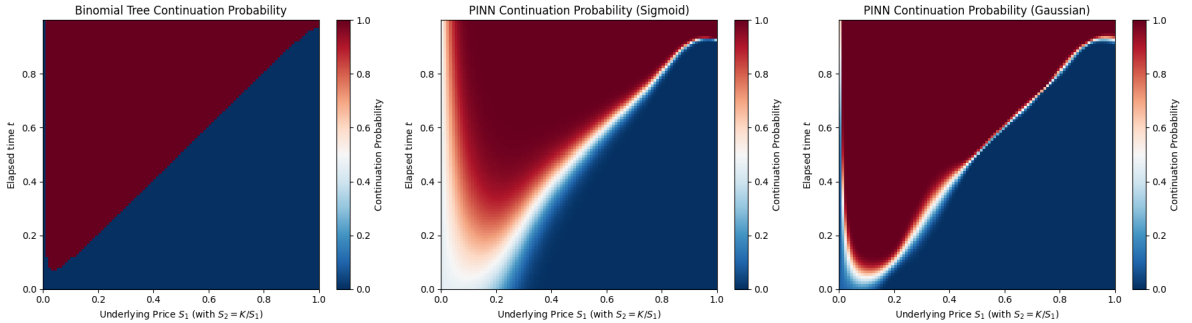


Figure 3.16 Estimated continuation probabilities for 2-dimensional case on $S_1 = S_2$ cross section across time

3.6 Minimum of two assets

To demonstrate the flexibility of our PINN framework, we will now shift our focus onto different payoffs on multiple assets as an extension. [22] explores a variety of different payoffs for American options on multiple assets, but that with a particularly interesting free boundary is the one of the call on the maximum of two assets. We will deal with the analogous case for the put on the minimum of two assets.

To price this, we can simply change the payoff function in our loss function, and update the boundary conditions accordingly to the new payoff function, but the variational inequality and the rest of the framework remain unchanged. For the American put on the minimum of two

assets, the payoff function is given by $g(S_1, S_2) = (K - \min(S_1, S_2))^+$, and thus we must have the terminal boundary condition

$$v(T, S_1, S_2) = (K - \min(S_1, S_2))^+ \quad (3.86)$$

When one of the assets is very high (i.e., $S_1 \rightarrow \infty$ or $S_2 \rightarrow \infty$), the minimum of the two assets is just the other asset, and thus we have

$$v(t, \infty, S_2) = v_{1D}(t, S_2) \quad v(t, S_1, \infty) = v_{1D}(t, S_1) \quad (3.87)$$

where v_{1D} is the price of the one-dimensional American put option on a single asset, which we can compute with the different methods we have already explored. Finally, when one of the assets is worthless, the minimum of the two assets is just the worthless asset, and hence the value will be K as we should exercise immediately. Thus, the boundary condition would be

$$v(t, S_1, S_2) = K \quad \text{for } S_1 = 0 \text{ or } S_2 = 0 \quad (3.88)$$

Together, these boundary conditions (alongside the variational inequality which remains the same with the operator $\mathcal{L}$ as in 3.58) form the PINN framework which we can train.

In Figure 3.17, we can see that our PINN yields results that are in line with our expectations. We expect to see some symmetry along the $S_1 = S_2$ line as the payoff is symmetric in the two assets. I introduce some asymmetry by training the network where S_2 has a higher constant volatility than S_1 .

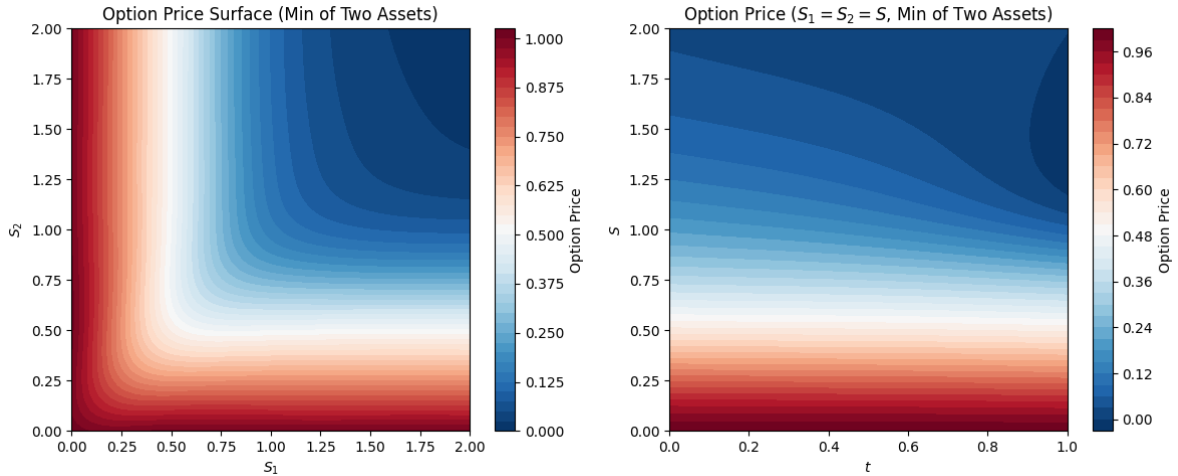


Figure 3.17 Heatmap for the put on the minimum of two assets across the S_1 - S_2 cross section at $t = 0$, and the S_1 - t cross section with $S_1 = S_2$

We can also see that the term structure is somewhat captured by the contour lines of the right figure of Figure 3.17, where the price decays as we approach maturity. I train the model with a high volatility of $(\sigma_1, \sigma_2) = (0.4, 0.5)$ to make the theta decay more significant. This also gives us a clearer evolution of the free boundary, which we see in Figure 3.18, where we can see that not only the continuation region is a mirror image of the theoretical result described in [22] as we would expect, but the PINN also captures the shrinking nature of the continuation region as we approach maturity, consistent with the intuition that the option loses time value and the region would close entirely, leaving only the box $\{(S_1, S_2) : S_1 \geq K, S_2 \geq K\}$ as the continuation region at maturity.

Overall, this demonstrates the capacity of our PINN approach to pricing options and we can see that the framework is flexible enough to be applied to different payoffs with minimal adjustments, yielding convincing results that are consistent with theoretical results. For the next

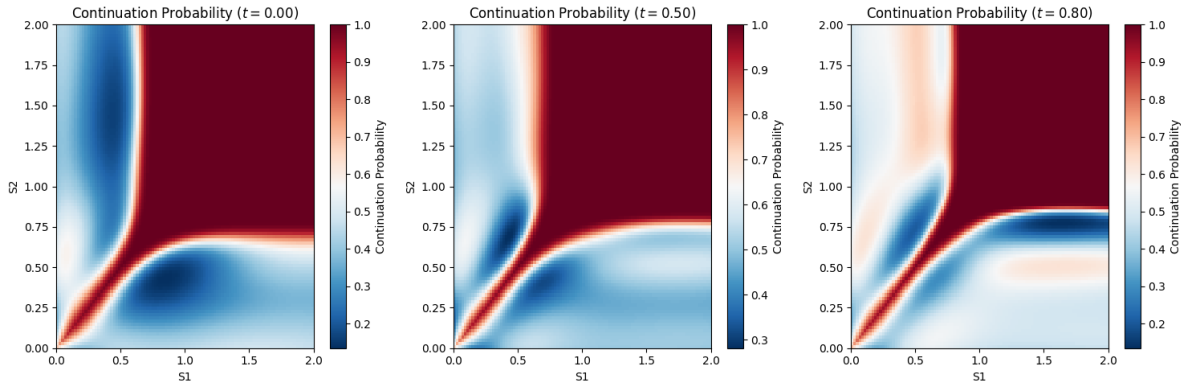


Figure 3.18 Continuation probabilities for the put on the minimum of two assets across the S_1 - S_2 cross section with varying t

step, we will move onto a different model, the Heston model, which injects randomness into the volatility, which is a more realistic assumption.

Chapter 4

The Heston Model

The classic Black Scholes model makes the strong assumption that the stock returns are normally distributed with known mean and variance. However, the Black Scholes formula does not depend on the mean, so allowing the mean to vary does not help us to generalise the model. But if we allow volatility to vary according to a stochastic process, we obtain a more generalised model. Heston [2] proposes a model that is not based on the Black Scholes formula that provides a closed-form solution for European call options. In this model, the dynamics under the pricing measure of the asset price S and the variance (squared volatility) process V are governed by the stochastic differential equation system under the risk-neutral measure $\mathbb{Q}$:

$$dS_t = rS_t dt + \sqrt{V_t} dW_t^1 \quad (4.1)$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t^2 \quad (4.2)$$

where we have initial conditions $S_0 = s > 0$ and $V_0 = v \geq 0$, and the two Brownian motions are correlated with

$$d\langle W^1, W^2 \rangle_t = \rho dt \quad \rho \in (-1, 1) \quad (4.3)$$

The dynamics of the variance V follow a CIR process with mean reversion rate $\kappa > 0$, long run state $\theta > 0$ and volatility of the volatility $\sigma > 0$. We can enforce the Feller condition $2\kappa\theta \geq \sigma^2$ so that the volatility process stays positive.

4.1 Closed-form Solution

There exists a closed-form solution derived by Heston [2] to the price of the European call option. This will serve as a benchmark for our other techniques, which would give us confidence in the accuracy of our other techniques for European options, and by extension, for American options too.

The closed-form solution is derived by using an Ansatz inspired by the Black-Scholes formula, and determining the elements of our guess by substituting it into the PDE we know the solution must satisfy along with the boundary conditions. We will include a sketch of the derivation here which follows those found in [2, 23], but with some further justification for a better overall understanding.

We will derive this PDE when we derive the variational equation in Section 4.3.1, but it is also an application of the Feynman-Kac formula, stating that the price of the European call option with strike K and maturity T must satisfy

$$\frac{\partial C}{\partial t} + rS \frac{\partial C}{\partial S} + \kappa(\theta - V) \frac{\partial C}{\partial V} + \frac{1}{2} V S^2 \frac{\partial^2 C}{\partial S^2} + \rho \sigma V S \frac{\partial^2 C}{\partial S \partial V} + \frac{1}{2} \sigma^2 V \frac{\partial^2 C}{\partial V^2} - rC = 0 \quad (4.4)$$

with the terminal condition $C(T, S, V) = (S - K)^+$.

We will mimic the form of the Black-Scholes formula to form the Ansatz for the solution

$$C(S, V, t) = SP_1(x, V, t) - Ke^{-r(T-t)}P_2(x, V, t), \quad x := \ln S \quad (4.5)$$

with terminal conditions $P_j(x, V, T) = \mathbf{1}_{\{x \geq \ln K\}}$ so that our Ansatz reduces to $(S - K)^+$ at maturity. We can substitute the Ansatz into the PDE and equate the coefficients of S and $Ke^{-r(T-t)}$ to obtain two PDEs for P_1 and P_2 :

$$\frac{\partial P_j}{\partial t} + (r + u_j V) \frac{\partial P_j}{\partial x} + (a - b_j V) \frac{\partial P_j}{\partial V} + \frac{1}{2} V \frac{\partial^2 P_j}{\partial x^2} + \rho \sigma V \frac{\partial^2 P_j}{\partial x \partial V} + \frac{1}{2} \sigma^2 V \frac{\partial^2 P_j}{\partial V^2} = 0 \quad (4.6)$$

where $u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $a = \kappa\theta$, $b_1 = \kappa - \rho\sigma$ and $b_2 = \kappa$ [2]. Another interpretation is, by Feynman-Kac, that P_j is the probability $\mathbb{Q}_j(x_T \geq \ln K \mid \mathcal{F}_t)$ where $\mathbb{Q}_j$ is a measure with the same dynamics as $\mathbb{Q}$ but with the drift of the log stock price process $x_t = \ln S_t$ changed to $r + u_j V_t$. This is a change of measure that can be done with Girsanov's theorem [9].

With this in mind, let's define the characteristic functions

$$\phi_j(x, V, t; \omega) = \mathbb{E}^{\mathbb{Q}_j} [e^{i\omega x_T} \mid \mathcal{F}_t] \quad (4.7)$$

which admits the Ansatz

$$\phi_j(\omega) = \exp(C_j(\tau, \omega) + D_j(\tau, \omega)V + i\omega x), \quad \tau := T - t \quad (4.8)$$

since the coefficients in the PDE for P_j are independent of x (justifying the $e^{i\omega x}$ term) and affine in V (justifying the $e^{D_j V}$ term) [24].

We can insert this Ansatz into the PDE for P_j to get a system of ODEs for C_j and D_j :

$$\frac{\partial D_j}{\partial \tau} = \frac{1}{2} \sigma^2 D_j^2 + (\rho \sigma i \omega - b_j) D_j + u_j i \omega - \frac{1}{2} \omega^2, \quad \frac{\partial C_j}{\partial \tau} = r i \omega + a D_j \quad (4.9)$$

with the terminal conditions $C_j(0, \omega) = D_j(0, \omega) = 0$. There exists a closed-form solution to this system of ODEs, where we first define

$$d_j := \sqrt{(\rho \sigma i \omega - b_j)^2 - \sigma^2(2u_j i \omega - \omega^2)} \quad (4.10)$$

$$g_j := \frac{b_j - \rho \sigma i \omega - d_j}{b_j - \rho \sigma i \omega + d_j} \quad (4.11)$$

where we use the opposite sign convention compared to Heston's original formulation for better numerical stability (more precisely to avoid regions of numerical discontinuity in the complex plane) as suggested in [25]. This yields the solutions

$$D_j(\tau, \omega) = \frac{b_j - \rho \sigma i \omega - d_j}{\sigma^2} \cdot \frac{1 - e^{-d_j \tau}}{1 - g_j e^{-d_j \tau}} \quad (4.12)$$

$$C_j(\tau, \omega) = r i \omega \tau + \frac{a}{\sigma^2} \left[(b_j - \rho \sigma i \omega - d_j) \tau - 2 \ln \frac{1 - g_j e^{-d_j \tau}}{1 - g_j} \right] \quad (4.13)$$

By the Gil-Pelaez inversion theorem [26], we can retrieve P_j from its characteristic function ϕ_j :

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-i\omega \ln K} \phi_j(\omega)}{i\omega} \right] d\omega \quad (4.14)$$

which we can substitute back into our original Ansatz for the call option (4.5) to get the closed-form solution for the call option price.

4.2 Tree Solution

Since the closed-form solution is limited to European options, we would like to find ways to approximate solutions to the pricing problem so that we can eventually extend the case to American options, which has no closed-form solution. The first method we will explore is the tree method, where it is easy to modify to the American case by adding an additional comparison to the intrinsic value at each node. We will use the closed-form formula as a benchmark for the tree method for European options, to give us confidence in the solution the tree model provides for American options as it is a simple modification in the algorithm, which we can use as a benchmark for the neural networks we will explore later.

The idea will be similar to the binomial tree we explored earlier in the paper for the Black-Scholes model, but we now have two correlated stochastic processes in the Heston model: the asset price process and the variance process. As this tree is not recombining, generating paths for these processes and computing the option prices by backwards induction naively would lead to an exponential time complexity, which is not computationally feasible. Therefore, we will use a grid interpolation approach as in [4] so that the number of nodes grows quadratically in the number of time steps instead.

For the algorithm, we will work under the variance process V and log stock price process Z (which is a simple transformation from S using Itô's formula):

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t^1 \quad (4.15)$$

$$dZ_t = (r - \frac{1}{2}V_t)dt + \sqrt{V_t}dW_t^2 \quad (4.16)$$

where $d\langle W^1, W^2 \rangle_t = \rho$ under the risk-neutral measure $\mathbb{Q}$ and $\kappa, \sigma, \theta, r > 0$ as before. Of course, the value C of the European option with maturity T and payoff function in terms of stock and volatility values $g : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$ will be

$$C(t, V_t, Z_t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[g(e^{Z_t}, \sqrt{V_t}) \mid \mathcal{F}_t] \quad (4.17)$$

4.2.1 Euler scheme

To construct the tree, we need to discretise the process. Suppose we have n steps, then $\Delta t = \frac{T}{n}$. Using the standard Euler scheme, we can have

$$V_{k+1} = V_k + \kappa(\theta - V_k^+)\Delta t + Y_{k+1}^1\sigma\sqrt{V_k^+\Delta t} \quad (4.18)$$

$$Z_{k+1} = Z_k + (r - \frac{1}{2}V_k)\Delta t + Y_{k+1}^2\sqrt{V_k^+\Delta t} \quad (4.19)$$

where the increment variables (Y_k^1, Y_k^2) are iid in k with probabilities

$$\mathbb{Q}^n(Y_k^1 = i, Y_k^2 = j) = \frac{1}{4}(1 + ij\rho) \quad i, j \in \{-1, 1\} \quad (4.20)$$

We use a different measure $\mathbb{Q}^n$ as it is not the same as the risk-neutral measure in the Heston model. To ensure positivity of the process, we have taken the positive part of the variance process $V_k^+ := \max(0, V_k)$.

4.2.2 Interpolation optimisation

The problem with the naïve Euler scheme is that the tree is not recombining, and hence the number of nodes scales exponentially with increasing time steps, which is not feasible computationally. We will hence, for each time step k , construct a grid of option prices at time k and the backward induction would be a bilinear interpolation of the four nearest points.

To set this up, we need the boundaries of the grid at each time step k , which is given by the minimum and maximum values of V_k and Z_k in the support. In other words, we have

$$z_k^{\max} = \max\{z : \mathbb{Q}^n(Z_k = z) > 0\} \quad (4.21)$$

$$z_k^{\min} = \min\{z : \mathbb{Q}^n(Z_k = z) > 0\} \quad (4.22)$$

$$v_k^{\max} = \max\{v : \mathbb{Q}^n(V_k = v) > 0\} \quad (4.23)$$

$$v_k^{\min} = \min\{v : \mathbb{Q}^n(V_k = v) > 0\} \quad (4.24)$$

$$(4.25)$$

With these boundaries, we have the grid spacing given our mesh sizes $m_z, m_v \in \mathbb{N}^+$:

$$\Delta z_k = (z_k^{\max} - z_k^{\min})/m_z \quad (4.26)$$

$$\Delta v_k = (v_k^{\max} - v_k^{\min})/m_v \quad (4.27)$$

and the gridpoints:

$$\hat{\mathcal{S}}_k = \left\{ \left(v_k^{\min} + i\Delta v_k, z_k^{\min} + j\Delta z_k \right) \mid i \in \{0, \dots, m_v\}, j \in \{0, \dots, m_z\} \right\}$$

We will now define the discretised process $(\tilde{V}, \tilde{Z})$ on the grid space $\hat{\mathcal{S}}_k$. This will involve the Euler scheme functions

$$\nu(y, v, z) = v + \kappa(\theta - v^+)\Delta t + y\sigma\sqrt{v^+\Delta t} \quad (4.28)$$

$$\zeta(y, v, z) = z + (r - \frac{1}{2}v)\Delta t + y\sqrt{v^+\Delta t} \quad (4.29)$$

Then using these functions, we can retrieve the next step. We can evolve our process to the next step with

$$(\tilde{V}_{k+1}, \tilde{Z}_{k+1}) = \text{snap}_{Y_{k+1}^3, Y_{k+1}^4}^{\hat{\mathcal{S}}_{k+1}} (\nu(Y_{k+1}^1, \tilde{V}_k, \tilde{Z}_k), \zeta(Y_{k+1}^1, \tilde{V}_k, \tilde{Z}_k)) \quad (4.30)$$

Where $\text{snap}_{i_3, i_4}^{\mathcal{S}} : \mathbb{R}^2 \rightarrow \mathcal{S} \subset \mathbb{R}^2$ is a function mapping $(v, z) \in \mathbb{R}^2$ to one of the grid points in $\mathcal{S} := \mathcal{S}_v \times \mathcal{S}_z$. $i_3, i_4 \in \{0, 1\}$ are additional parameters that determine the corner in which (v, z) is mapped. Visually, this is illustrated in Figure 4.1, where each point in the price-variance space can evolve to four different points in the next time step, each of which we take the interpolation of the four nearest grid points.

To do this, we define the 1D snapping functions $\phi_i^{\mathcal{T}} : \mathbb{R} \rightarrow \mathcal{T}$ where we have the one-dimensional grid points $\mathcal{T} \subset \mathbb{R}$ and the snap-right indicator $i \in \{0, 1\}$. The function is defined mathematically as

$$\phi_0^{\mathcal{T}}(u) = \max\{u' \in \mathcal{T} : u' \leq u\} \quad (4.31)$$

$$\phi_1^{\mathcal{T}}(u) = \min\{u' \in \mathcal{T} : u' \geq u\} \quad (4.32)$$

Our 2D snapping function $\text{snap}_{i_3, i_4}^{\mathcal{S}} : \mathbb{R}^2 \rightarrow \mathcal{S}$ where $\mathcal{S} = \mathcal{S}_v \times \mathcal{S}_z \subset \mathbb{R}^2$ is thus

$$(v, z) \mapsto (\phi_{i_3}^{\mathcal{S}_v}(v), \phi_{i_4}^{\mathcal{S}_z}(z)) \quad (4.33)$$

The joint probabilities [4] for Y_{k+1} is given by, for $i_1, i_2 \in \{-1, 1\}, i_3, i_4 \in \{0, 1\}$:

$$q_{i_1, i_2, i_3, i_4}(\tilde{V}_k, \tilde{Z}_k) := \tilde{\mathbb{Q}}(Y_{k+1}^1 = i_1, Y_{k+1}^2 = i_2, Y_{k+1}^3 = i_3, Y_{k+1}^4 = i_4 \mid (\tilde{V}_k, \tilde{Z}_k)) \quad (4.34)$$

$$= \frac{1}{4} (1 + i_1 i_2 \rho) \times c_{i_3 i_4}^{\hat{\mathcal{S}}_{k+1}} \left(\nu(i_1, \tilde{V}_k, \tilde{Z}_k), \zeta(i_2, \tilde{V}_k, \tilde{Z}_k) \right) \quad (4.35)$$

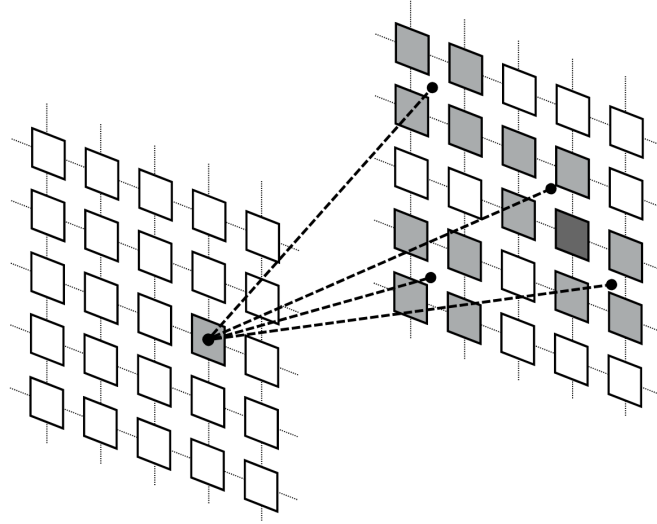


Figure 4.1 Interpolation illustration (adapted from [4])

Here we have used the interpolation quantity [27], where on a general grid set $\mathcal{S} = S_v \times S_z \subset \mathbb{R}^2$ we have

$$c_{i_3 i_4}^{\mathcal{S}}(v, z) = \begin{cases} (1 - \tilde{v})(1 - \tilde{z}) & (i_3, i_4) = (0, 0) \\ \tilde{v}(1 - \tilde{z}) & (i_3, i_4) = (1, 0) \\ (1 - \tilde{v})\tilde{z} & (i_3, i_4) = (0, 1) \\ \tilde{v}\tilde{z} & (i_3, i_4) = (1, 1) \end{cases}$$

where we have the difference terms

$$\tilde{v} = \frac{v - \phi_0^{\mathcal{S}_v}(v)}{\phi_1^{\mathcal{S}_v}(v) - \phi_0^{\mathcal{S}_v}(v)} = \frac{v - \phi_0^{\mathcal{S}_v}(v)}{\Delta v_k} \quad (4.36)$$

$$\tilde{z} = \frac{z - \phi_0^{\mathcal{S}_z}(z)}{\phi_1^{\mathcal{S}_z}(z) - \phi_0^{\mathcal{S}_z}(z)} = \frac{z - \phi_0^{\mathcal{S}_z}(z)}{\Delta z_k} \quad (4.37)$$

because our grid is evenly spaced at all time steps k .

4.2.3 Backwards induction

We can now work backwards on the tree, having constructed the grids. Our base case is the final grid, where the price is just the payoff as we are at maturity. For a general time t_k at step k and node $(v, z) \in \hat{\mathcal{S}}_k$, we will generate the four possible steps determined by y_1, y_2 each taking values in $\{-1, 1\}$, and evolve our node with

$$(v, z) \mapsto (\nu(y_1, v, z), \zeta(y_2, v, z)) =: (v', z') \quad (4.38)$$

Which will be in between four points on the grid $\hat{\mathcal{S}}_{k+1}$. These can be found by performing the snap function on our evolved node: for $y_3, y_4 \in \{0, 1\}$ we can encapsulate $y = (y_1, y_2, y_3, y_4)$ and we have

$$(v_y, z_y) = \text{snap}_{y_3, y_4}^{\hat{\mathcal{S}}_{k+1}}(v', z') \quad (4.39)$$

Each of these will have a known option price. Therefore, we can just compute the discounted expectation of these sixteen nodes (two choices for each of $y_1, \dots, y_4$) for the option price at

the original node:

$$C(t_k, v, z) = e^{-r\Delta t} \sum_{y \in \mathcal{Y}} q_{y_1, y_2, y_3, y_4}(v, z) \times C(t_{k+1}, v_y, z_y) \quad (4.40)$$

where $\mathcal{Y} = \{-1, 1\}^2 \times \{0, 1\}^2$.

4.2.4 Implementation techniques

If we implemented this method naively, it would be very difficult for us to use the binomial tree as a benchmark because it would need to construct a new tree from scratch for every single value of S_0 , V_0 and T . Thus any heatmaps or plots we create would require hundreds of trees which is not feasible, considering each tree requires many time steps and large mesh sizes to be accurate.

Thus in addition to vectorising our code for each tree, we will instead consider an initial range $[S_0^{\min}, S_0^{\max}]$ and $[V_0^{\min}, V_0^{\max}]$ which we will use to construct our grid bounds. More specifically, $z_k^{\max}$ and $v_k^{\max}$ will be the up-most path starting with $(S_0^{\max}, V_0^{\max})$ instead of simply some individual value (S_0, V_0) , and similar for the down paths. In this way, when we use the binomial tree as a benchmark, we can use any initial value of (S_0, V_0) as long as it's in the original range and compute the price with an interpolation as described above. With this method, we can evaluate across different initial values while constructing just one tree. We sacrifice a little accuracy in doing so because the grids would be more widely spaced, which we can combat by increasing the mesh sizes. Thus, we effectively construct one more complex tree instead of hundreds of simpler ones, which is much more computationally efficient.

As for selecting our mesh sizes, we will use the result that if we have

$$\lim_{n \rightarrow \infty} \Delta t = 0, \quad \lim_{n \rightarrow \infty} \max_{k=1, \dots, n-1} \frac{\Delta v_k^n}{\Delta t_k} = 0, \quad \lim_{n \rightarrow \infty} \max_{k=1, \dots, n-1} \frac{\Delta z_k^n}{\Delta t_k} = 0 \quad (4.41)$$

then the process represented in our Heston tree converges weakly to the original process (V, Z) [4]. Note here we denote Δv_k from before as Δv_k^n to emphasize we have split our maturity T into n steps, and similarly for Δz_k^n .

Bearing this in mind, a resolution of $n = 100, m_v = 300, m_z = 600$ allows our tree to faithfully reproduce the closed-form solution, which are presented in the numerical results in section 4.5.1.

4.3 PINN Solution

4.3.1 Variational equation

We would now like to derive the variational equation for options on assets under the Heston model. Note that for a general process $X_t \in \mathbb{R}^d$ solving the Itô SDE

$$dX_t = \mathbf{b}(X_t)dt + \boldsymbol{\sigma}(X_t)d\mathbf{W}_t \quad (4.42)$$

where we have the drift $\mathbf{b} : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the volatility $\boldsymbol{\sigma} : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ and the m -dimensional Brownian motion $\mathbf{W}_t$. On second-order smooth functions $f \in C^2(\mathbb{R}^d)$, the infinitesimal generator $\mathcal{L}$ is defined as [28]

$$\mathcal{L}f(x) := \lim_{t \rightarrow 0} \frac{\mathbb{E}[f(X_t) \mid X_0 = x] - f(x)}{t} \quad (4.43)$$

which exists and can be derived with Itô's formula (or Taylor's expansion) to be [29]

$$\mathcal{L}f(x) = \sum_{i=1}^d \mathbf{b}_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d (\boldsymbol{\sigma} \boldsymbol{\sigma}^\top)_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (4.44)$$

Let's apply this to our Heston model. The drift vector is

$$\mathbf{b}(S, V) = \begin{pmatrix} rS \\ \kappa(\theta - V) \end{pmatrix} \quad (4.45)$$

To compute $\boldsymbol{\sigma} \boldsymbol{\sigma}^\top$, we need to represent our correlated Brownian motions in terms of independent ones. Suppose we have independent Brownian motions B^1 and B^2 , we can define

$$W^1 = B^1 \quad (4.46)$$

$$W^2 = \rho B^1 + \sqrt{1 - \rho^2} B^2 \quad (4.47)$$

which has the quadratic variation $d\langle W^1, W^2 \rangle_t = \rho dt$ as required. Using these new Brownian motions, we can rewrite the SDE as

$$d \begin{pmatrix} S_t \\ V_t \end{pmatrix} = \underbrace{\begin{pmatrix} rS_t \\ \kappa(\theta - V_t) \end{pmatrix}}_{\mathbf{b}(S_t, V_t)} dt + \underbrace{\begin{pmatrix} S_t \sqrt{V_t} & 0 \\ \rho \sigma \sqrt{V_t} & \sigma \sqrt{V_t} \sqrt{1 - \rho^2} \end{pmatrix}}_{\boldsymbol{\sigma}(S_t, V_t)} \begin{pmatrix} dB_t^1 \\ dB_t^2 \end{pmatrix} \quad (4.48)$$

which means that our covariance matrix is now

$$\boldsymbol{\sigma}(S, V) = \begin{pmatrix} S\sqrt{V} & 0 \\ \rho\sigma\sqrt{V} & \sigma\sqrt{V}\sqrt{1 - \rho^2} \end{pmatrix}. \quad (4.49)$$

which means

$$\boldsymbol{\sigma} \boldsymbol{\sigma}^\top = \begin{pmatrix} S^2 V & \rho\sigma S V \\ \rho\sigma S V & \sigma^2 V \end{pmatrix} \quad (4.50)$$

We now have everything we need for the formula. For any function $f = f(S, V) \in C^2(\mathbb{R}^d)$, we have

$$\mathcal{L}f = rS \frac{\partial f}{\partial S} + \kappa(\theta - V) \frac{\partial f}{\partial V} + \frac{1}{2} \left(S^2 V \frac{\partial^2 f}{\partial S^2} + 2\rho\sigma S V \frac{\partial^2 f}{\partial S \partial V} + \sigma^2 V \frac{\partial^2 f}{\partial V^2} \right) \quad (4.51)$$

This is our infinitesimal generator $\mathcal{L}$ acting on smooth functions f . Suppose we have a European option with payoff $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, then our price function $u(t, S, V)$, the value of the call option with expiry T at time t with $S_t = S$ and $V_t = V$ satisfies

$$\mathcal{G}[u] := ru - \frac{\partial u}{\partial t} + \mathcal{L}u = 0 \quad (4.52)$$

with terminal condition $u(T, S, V) = g(S)$. This matches the PDE we had stated in the closed-form solution in section 4.4.

As a check, we can set $\kappa = 0$ and $\sigma = 0$ so that $dV = 0$ and volatility becomes a constant, and our model collapses to what we have been working with before. Our generator becomes

$$\mathcal{L}f = rS \frac{\partial f}{\partial S} + \frac{1}{2} S^2 V_0 \frac{\partial^2 f}{\partial S^2} \quad (4.53)$$

which matches our expression (3.35) from the Black-Scholes section, since V_0 is the initial (and hence constant) variance.

We have derived the variational equation for the European option for simplicity, but it can be easily extended for the American case. With the same argument as (3.37), the value of the American option must satisfy the PDE

$$\min(\mathcal{G}[u], v(t, S, V) - g(S)) = 0 \quad (4.54)$$

4.3.2 PINN formulation

We can fit the variational equation (4.54) into the PINN framework established in section 2.3 which will serve as the PDE loss (2.36), and all that is left is to set up the loss function and the relevant boundary conditions. It will be very similar to the Black-Scholes case, but as we have an additional dimension V for the variance, we will need additional boundary conditions to ensure regularity in the variance dimension as well. These boundary conditions will be inspired by those given for the European call option in [2], but we will adapt them in our study, as they exhibit different behaviour for American put options.

1. $t \rightarrow T$: the value at expiry is just the intrinsic value, or the payoff of the put option:

$$v(T, S, V) = \max(0, K - S) \quad (4.55)$$

which will form the terminal loss (2.37) for the PINN.

2. $S \rightarrow 0$: the option value is the strike when the stock is worthless, because we can exercise immediately (for European options we would require a discounting term):

$$v(t, 0, V) = K \quad (4.56)$$

3. $S \rightarrow \infty$: the value of a very far out of the money option is 0, as it is very unlikely for it to expire in of the money:

$$v(t, \infty, V) = 0 \quad (4.57)$$

4. $V \rightarrow 0$: volatility being zero is a slightly more complicated case as Heston proposes a collapsed first-order PDE, which does not yield good results in practice. Instead, we will set the condition to be the payoff under the guise that asset price becomes deterministic, which is only true instantaneously due to the mean-reverting nature of the variance process, but it is a good approximation for small volatilities:

$$v(t, S, 0) = \max(0, K - S) \quad (4.58)$$

5. $V \rightarrow \infty$: when the volatility is very high, it is almost sure that the product of assets would be zero at some point before expiry, so the value of the option is the strike:

$$v(t, S, \infty) = K \quad (4.59)$$

In practice, as we clearly cannot represent infinity, we will settle for large constants for the asset price S_{inf} and variance V_{inf} . The minimum for both of these will be taken to be zero. Thus our domain is

$$\Omega = (0, T) \times (0, S_{\text{max}}) \times (0, V_{\text{max}}) \quad (4.60)$$

The boundary conditions (2)-(5) together will form the boundary loss (2.38) for the PINN. By choosing relevant weights for each of these loss components (we could additionally set different values for different individual boundary components), we can ensure that the neural network learns the correct behaviour at the boundaries, which is crucial for the accuracy of the solution.

4.4 Multi-asset PINN

With the neural network established, we are now ready to move on to pricing options on multiple assets. As with the Black-Scholes case, we will price the put on the product of multiple assets because we can find a one-dimensional reduction. Here, we will consider a simplified model. Suppose we have n assets and our assets $S = (S_1, \dots, S_n)^\top$ are governed by the stochastic differential equations:

$$dS = \text{diag}(S)(r dt + \sqrt{V}\sigma\Sigma dW_1) \quad (4.61)$$

$$dV = \kappa(\theta - V) dt + \bar{\sigma}\sqrt{V} dW_2 \quad (4.62)$$

where we have the vector of iid Brownian motions $W_1 = (W_{1,1}, \dots, W_{1,n})^\top$, variance vector $\sigma = (\sigma_1, \dots, \sigma_n)^\top$ and the covariance matrix Σ , where for simplicity we will assume all cross-correlations $\Sigma_{ij} = \rho$ for $i \neq j$. Assume additionally that each asset is correlated with the variance process: $dW_{1,j}dW_2 = \rho_j dt$. Note that this means the asset S_i satisfies

$$dS_i = S_i \left(r dt + \sqrt{V}\sigma_i \sum_{j=1}^n \Sigma_{ij} dW_{1,j} \right) \quad (4.63)$$

To guarantee positiveness of the variance process, we will enforce the Feller condition on the parameters

$$2\kappa\theta \geq \bar{\sigma}^2 \quad (4.64)$$

We make these assumptions to allow a one-dimensional reduction for the Heston model, which would provide us a comparable benchmark for our results with the tree method. This would in turn give us confidence for our approximated solutions, should we wish to relax assumptions or to change the payoff.

4.4.1 One-dimensional reduction

Since we would like to find an equivalent formulation of the product of assets $X := \prod_{i=1}^n S_i$, it would be easier to work in the log domain. That is, find the governing SDE for the log process $Y = \log X = \sum_{i=1}^n \log S_i$ and then exponentiate with Itô's lemma to recover the desired result.

Let $Y_i = \log S_i$. Then by Itô's lemma, we have

$$dY_i = \frac{1}{S_i} dS_i - \frac{1}{2S_i} dS_i dS_i \quad (4.65)$$

$$= r dt + \sqrt{V}\sigma_i \sum_{j=1}^n \Sigma_{ij} dW_{1,j} - \frac{1}{2} V \sigma_i^2 dt \quad (4.66)$$

because cross-correlations are zero within W_1 . We can collect terms to get

$$dY = \left(nr - \frac{1}{2} V \sum_{i=1}^n \sigma_i^2 \right) dt + \sqrt{V} \sum_{j=1}^n \sum_{i=1}^n \sigma_i \Sigma_{ij} dW_{1,j} \quad (4.67)$$

For ease of notation, let $A = \sum_{i=1}^n \sigma_i^2$ and $b_j = \sum_{i=1}^n \sigma_i \Sigma_{ij}$. Then we have

$$dY = \left(nr - \frac{1}{2} V A \right) dt + \sqrt{V} \sum_{j=1}^n b_j dW_{1,j} \quad (4.68)$$

Now Y has variance

$$dY dY = V \sum_{j=1}^n b_j^2 dt =: VB dt \quad (4.69)$$

where we let $B := \sum_{j=1}^n b_j^2$ for convenience, we can rewrite dY with a new Brownian motion:

$$dY = \left(nr - \frac{1}{2}VA \right) dt + \sqrt{VB} dW_X \quad (4.70)$$

where we have

$$dW_X = \frac{\sum_{j=1}^n b_j dW_{1,j}}{\sqrt{B}} \quad (4.71)$$

Using our correlation structure, we have

$$b_j = \sigma_j + \rho \sum_{i \neq j} \sigma_i = (1 - \rho)\sigma_j + \rho \underbrace{\sum_{i=1}^n \sigma_i}_{=: S_\sigma} \quad (4.72)$$

So the sum of squares B is just

$$B = \sum_{j=1}^n b_j^2 = \sum_{j=1}^n ((1 - \rho)\sigma_j + \rho S_\sigma)^2 \quad (4.73)$$

$$= (1 - \rho)^2 A + S_\sigma^2 (2\rho(1 - \rho) + n\rho^2) \quad (4.74)$$

We now have all components to exponentiate to recover $X = e^Y$. Using Itô's lemma again, we have

$$dX = X dY + \frac{1}{2} X dY dY \quad (4.75)$$

$$= X \left(nr + \frac{1}{2} V(B - A) \right) dt + X \sqrt{VB} dW_X \quad (4.76)$$

This is close to the Heston model, with the diffusion $\tilde{V} = VB$ which gives us

$$d\tilde{V} = B dV \quad (4.77)$$

$$= \kappa(B\theta - \tilde{V}) dt + \bar{\sigma} \sqrt{B} \sqrt{\tilde{V}} dW_2 \quad (4.78)$$

which is a new CIR process $\tilde{V}$ with parameters $\tilde{\kappa} = \kappa$, $\tilde{\theta} = B\theta$ and $\tilde{\sigma} = \bar{\sigma}\sqrt{B}$. The product of assets nearly gives a direct reduction to a one-dimensional Heston process, but the drift of X contains V which is a stochastic process itself. Thus for simplicity and for our benchmark, we will set $\rho = 0$ which means $B = A$ and the drift term becomes deterministic. As such, we are able to recover a one-dimensional Heston model for the product of assets X with variance $\tilde{V}$ and drift nr .

4.4.2 Variational equation

Recall for a process $X_t \in \mathbb{R}^d$ solving $dX_t = b(X_t)dt + \sigma(X_t)dW_t$, the infinitesimal generator $\mathcal{L}$ can be derived with Itô's formula [29] on $f(x)$ to be

$$\mathcal{L}f(x) = \sum_{i=1}^d b_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \left(\sigma \sigma^\top \right)_{ij} (x) \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (4.79)$$

In our case, we have a function $f = f(S_1, \dots, S_d, V)$, and we can derive the infinitesimal generator much like the 1D case, but this time we need to sum over our different asset and variance processes. Mathematically, our generator can be expressed to be

$$\mathcal{L}f = \sum_{i=1}^d r S_i \frac{\partial f}{\partial S_i} + \kappa(\theta - V) \frac{\partial f}{\partial V} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \Sigma_{ij} V S_i S_j \frac{\partial^2 f}{\partial S_i \partial S_j} \quad (4.80)$$

$$+ \frac{1}{2} \bar{\sigma}^2 V \frac{\partial^2 f}{\partial V^2} + \sum_{i=1}^d \rho_i \sigma_i \bar{\sigma} V S_i \frac{\partial^2 f}{\partial S_i \partial V} \quad (4.81)$$

For any payoff $g(\cdot)$, we have, as before, the variational equation for an European option being

$$\mathcal{G}[f] := rf - \frac{\partial f}{\partial t} + \mathcal{L}f = 0 \quad (4.82)$$

and for an American option, we can introduce the early stopping criterion and the variation equation becomes

$$\min(\mathcal{G}[f], f - g) = 0 \quad (4.83)$$

4.4.3 PINN formulation

This will be very similar to the one-dimensional case, but we will make a few modifications to account for the additional dimensions. The PDE loss will be the same as before, but the boundary conditions will include:

1. $t \rightarrow T$: the value at expiry is just the intrinsic value which is the payoff:

$$u(T, S_1, \dots, S_n, V) = \max \left(0, K - \prod_{k=1}^n S_k \right) \quad (4.84)$$

which serves as our terminal loss (2.37) for the PINN.

2. $S \rightarrow 0$: if one of the assets reaches zero, then the product of assets is consequently zero and we should exercise immediately. Thus the value is the strike. To implement this, let $\tilde{S}$ be S with one of the assets modified manually to 0. Then we have

$$v(t, \tilde{S}, V) = K \quad (4.85)$$

3. $S \rightarrow \infty$: if one of the assets has a very high price, then the product of assets would also be very high meaning the option would expire worthless out of the money. To implement this, let $\bar{S}$ be S with one of the assets modified manually to S_{inf} , a large constant. Then the boundary condition is

$$v(t, \bar{S}, V) = 0 \quad (4.86)$$

4. $V \rightarrow 0$: with the same logic as the one-dimensional case, we will set the price under zero volatility to be zero. Hence we have

$$v(t, S, 0) = \max \left(0, K - \prod_{k=1}^n S_k \right) \quad (4.87)$$

5. $V \rightarrow \infty$: with infinite volatility, it is almost sure that the intrinsic value of the option would reach the strike at some time before expiry, and thus the value of the option is the strike for the put. To implement this, set V_{inf} a large constant, and we have

$$v(t, S, V_{\text{inf}}) = K \quad (4.88)$$

4.5 Numerical Results

4.5.1 Interpolating tree method

As mentioned before, we choose the resolution $n = 100, m_v = 300, m_z = 600$ which allows a faithful approximation of closed form solution while keeping runtime low. Figure 4.2 are the results after using the parameters $K = 1, T = 1, r = 0.05, \kappa = 2, \theta = 0.04, \sigma = 0.3, \rho = -0.7$. We produce these plots by varying across different values of S_0 (for delta), v_0 (for vega) and time elapsed (for term structure), and all else kept constant.

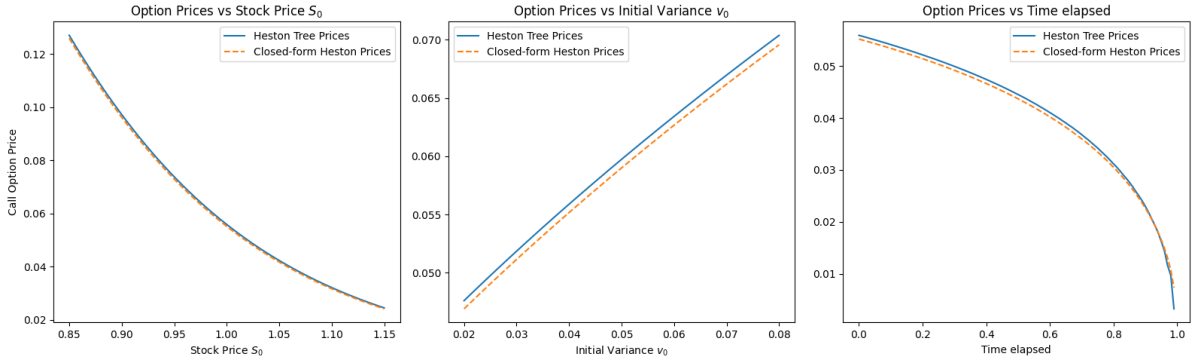


Figure 4.2 Call option prices by the Heston Tree

We can see that the Heston tree does indeed replicate the closed form solution very closely to very small error. This gives us confidence in our implementation, and with an easy modification we can compute the American option prices which we will use as a benchmark.

To compute the American option price with the tree model, all we need to do is to introduce the stopping condition at every node: we evaluate the intrinsic value of the option at each node and set the option value to be the maximum of the intrinsic value and the discounted continuation value.



Figure 4.3 American vs European option prices for the put

In Figure 4.3 we can see that the European option price indeed bounds the American prices from below, as we would expect. We have confidence that our American prices are accurate since our binomial tree faithfully approximates the closed-form solution for European options. We will now use this as a benchmark for neural network approximations.

4.5.2 One-dimensional PINN solution

We are now ready to train the neural network to price our option. We choose to use a learning rate of 1×10^{-3} with the Adam optimiser and a scheduler with step size 2000 and gamma 0.7 and the Sigmoid activation function. By using parameters $K = 1, T = 1, r = 0.1, \kappa = 2, \theta = 0.04, \sigma = 0.3, \rho = -0.7$ obtain the plots as shown in Figure 4.4 for the American put.

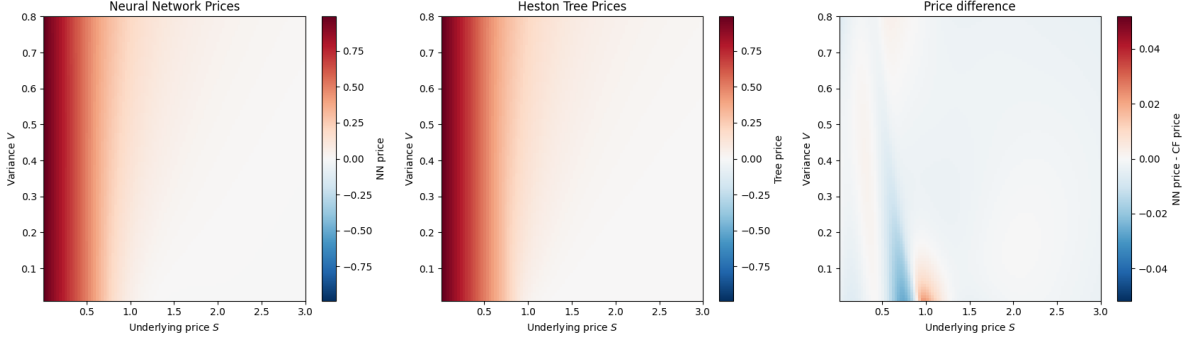


Figure 4.4 American put option prices by the PINN and the tree across S and V parameters at time $t = 0$

We can see that the neural network indeed faithfully reproduces the solution approximated by the binomial tree, even though it tends to perform less well at the money especially at lower volatilities. We can take slices of our three dimensional heatmap on the triple (t, S, V) to get a better understanding of the performance, by fixing two of the variables at the default $(t = 0, S = 1.0, V = 0.1)$ and plotting the option price against the other variable. This is shown in Figure 4.5.

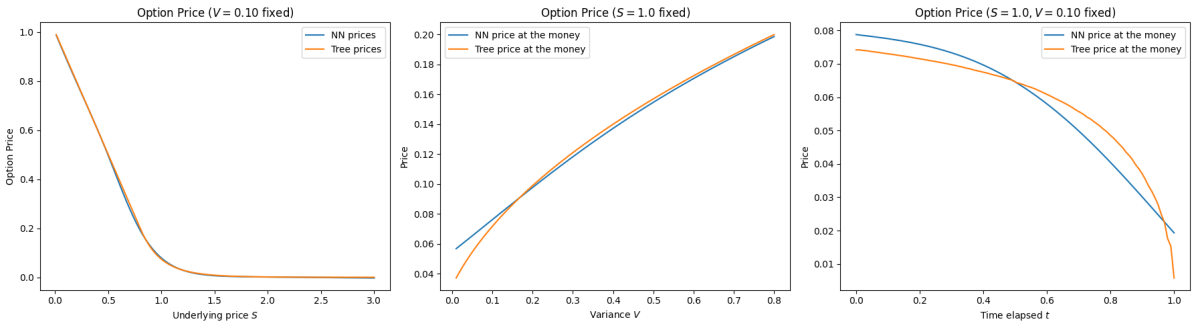


Figure 4.5 PINN vs tree prices varying across S , V and t parameters

We can see that the PINN does indeed capture well the general shape and behaviour of the option price, especially for varying across S and V but it struggles a bit more with the term-structure. That said, it is still able to capture the theta decay with its monotonicity, which is a good sign.

4.5.3 Multi-asset PINN solution

The multi-asset case is more difficult to train, as we have a higher dimensional input space and requires a greater batch size. I choose 10000 to be the batch size with the same learning rate and scheduler as before, but training takes several hours to complete and hence we will neglect taking an ensemble average and just present the results from one run. As discussed before, we will set our parameters satisfying the conditions required for a one-dimensional reduction. The

left figure of Figure 4.6 shows the heatmap of the option prices across the S_1 - S_2 plane with the time fixed at $t = 0$ and initial variance fixed at $V = V_0 = 0.04$.

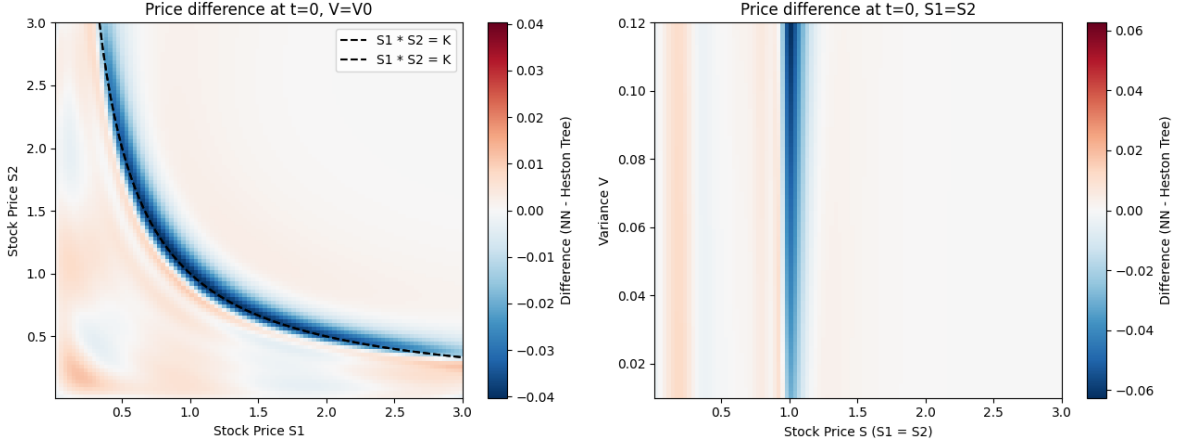


Figure 4.6 Heatmap of the difference between PINN and tree prices

We can see that the fit is good in the money and out of the money but struggles much more at the money, which is expected as the option price is more sensitive, and we are now in a four-dimensional input space and hence accuracy is more difficult to achieve. We will zoom in on the at the money region by fixing $t = 0$ and $S_1 = S_2 = \sqrt{K}$ and plotting the option price across V , which is shown in the right figure of Figure 4.6.

We can see that the weakness of the PINN is along the strike price with $S_1 = S_2 = \sqrt{K}$, and the accuracy is better for lower volatilities. Overall, the results are okay, given that we are sacrificing some accuracy in order to be able to price along the whole input domain, but we will expect the PINN to struggle more with the estimating free boundaries correctly since there is a bias in the prices already. However, the shape of the free boundary should still be correct, so there is a bit of value of estimating the free boundaries, which we will do in the next section.

4.5.4 Free boundary

One-dimensional case

We will first present the free boundary for the one-dimensional case. As this is a boundary on three dimensions, we will present it as cross sections, where one variable will be fixed at their default value ($t = 0, S = 1.0, V = 0.04$) and the other two will be plotted against each other. We will omit the t - S plane as it will look very similar to the free boundary for the Black-Scholes case 3.13, and we will also omit the t - V plane as it is not very informative, with continuation probabilities close to 1 for the whole region near the money at initial time regardless of the initial variance. This leaves us the free boundary on the S - V plane, shown in Figure 4.7

We can see that the continuation region is reasonably well approximated with both methods, and we additionally observe some noise near the boundary which is due to the Monte Carlo estimation of the continuation value. Unfortunately, the network struggles to capture the curvature characteristic of the free boundary, which is unsurprising as it doesn't price the option with perfect accuracy in all areas. That said, the results are promising against the benchmark, even though we have a patch of the continuation probabilities that are closer to one further in the money (at around $(S, V) = (0.4, 0.3)$), which is due to the bias in the option price in the corresponding region when we had our price comparison heatmaps in Figure 4.4.

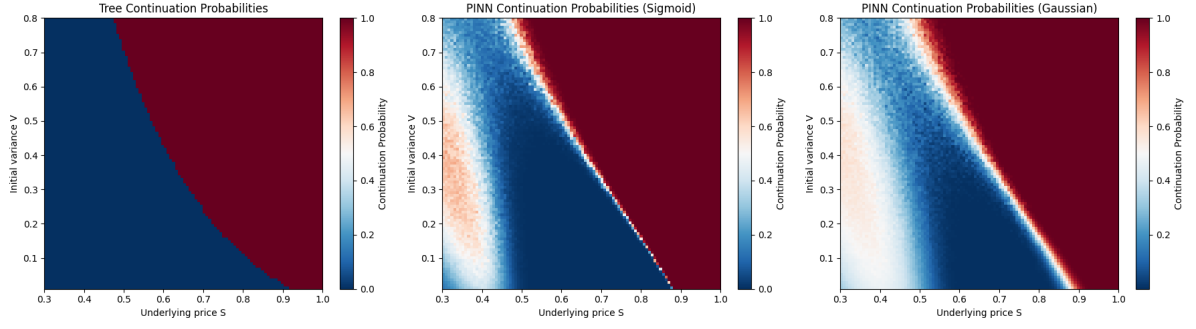


Figure 4.7 Continuation probabilities for the American put option on the S - V plane

Multi-asset case

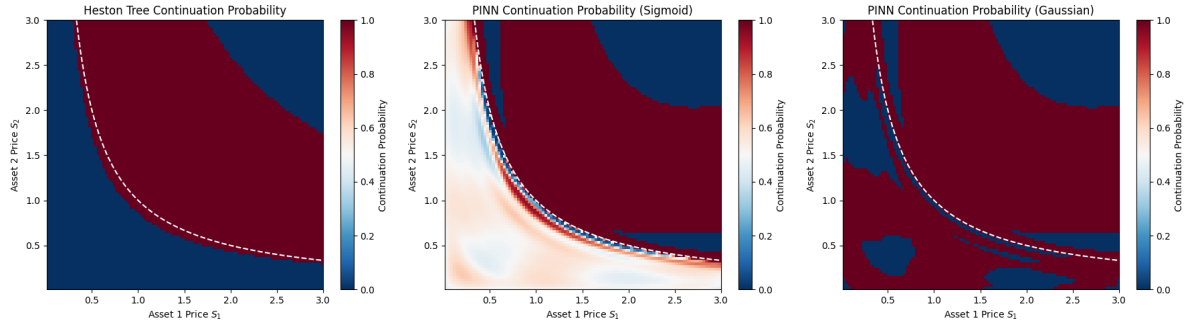


Figure 4.8 Continuation probabilities for the American put option on the S - V plane

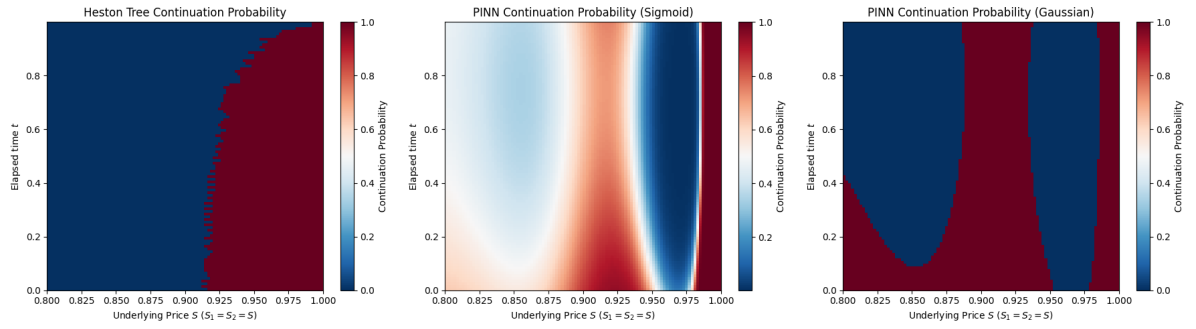


Figure 4.9 Continuation probabilities for the American put option on the S - t plane

Appendix A

Appendix 1

A.1 Change of measure for the Black-Scholes model

Consider the discounted stock price

$$\tilde{S}_t := \frac{S_t}{B_t} = e^{-rt} S_t \quad (\text{A.1})$$

and apply Itô's lemma to get a new SDE in terms of the discounted stock price

$$d\tilde{S}_t = \tilde{S}_t[(\mu - r)dt + \sigma dW_t^{\mathbb{P}}] \quad (\text{A.2})$$

which means that under the physical measure $\mathbb{P}$, $\tilde{S}_t$ has nonzero drift unless $\mu = r$, and hence not a martingale. So to enforce no arbitrage in our pricing model, we need a new equivalent probability measure, name it $\mathbb{Q}$, under which $\tilde{S}_t$ is a martingale.

Let's define the quantity

$$c := \frac{\mu - r}{\sigma} \quad (\text{A.3})$$

which measures how much excess drift per unit volatility the stock earns under the physical measure $\mathbb{P}$. Then we can define the Radon-Nikodym derivative

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = Z_t := \exp \left(-cW_t^{\mathbb{P}} - \frac{1}{2}c^2t \right) \quad (\text{A.4})$$

which satisfies Novikov's condition as c is constant so $\mathbb{Q}$ is well defined. The new process

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + ct \quad (\text{A.5})$$

is a Brownian motion under $\mathbb{Q}$ by Girsanov's theorem. If we substitute this into the original SDE, we obtain

$$\frac{dS_t}{S_t} = \mu dt + \sigma(W_t^{\mathbb{Q}} - cdt) \quad (\text{A.6})$$

$$= (\mu - \sigma c)dt + \sigma dW_t^{\mathbb{Q}} \quad (\text{A.7})$$

$$= rdt + \sigma dW_t^{\mathbb{Q}} \quad (\text{A.8})$$

So now under $\mathbb{Q}$, we have $d\tilde{S}_t = \sigma \tilde{S}_t dW_t^{\mathbb{Q}}$ which has zero drift, and hence $\tilde{S}_t$ is a $\mathbb{Q}$ -martingale. Intuitively, this change of measure reweights possible stock paths so that there is no more risk premium, and all assets earn the risk-free rate in expectation after discounting.

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